

CHARACTERISTIC CYCLES AND NON-ACYCLICITY CLASSES FOR CONSTRUCTIBLE ÉTALE SHEAVES

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To the memory of Professor Linsheng Yin (1963–2015)

ABSTRACT. Let $f : X \rightarrow Y$ be a morphism between smooth schemes over a perfect field k . Let $\mathcal{F}$ be a constructible sheaf on X and Z a closed subscheme of X such that f is $SS(\mathcal{F})$ -transversal outside Z . We construct a class supported on the non-transversal locus Z by using the characteristic cycle $CC(\mathcal{F})$ defined by T. Saito. This class is a geometric counterpart of the non-acyclicity class introduced by the third author and Zhao in [18]. Under certain conditions, the formation of this class is compatible with base change and proper pushforward. It also satisfies the Milnor formula and a conductor formula.

We conjecture that the image of this geometric class under the cycle class map is the non-acyclicity class. This conjecture can be viewed as a (relative version of) Milnor type formula for non-isolated singularities. We prove this conjecture when k is algebraically closed, X is quasi-affine and $Y = \mathbb{A}_k^1$.

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1. INTRODUCTION

1.1. Let k be a perfect field of characteristic $p > 0$ and $S = \operatorname{Spec} k$. Let Λ be a finite field of characteristic $\ell \neq p$. Let X be a smooth scheme over S and $f : X \rightarrow Y$ a flat morphism of finite type to a smooth curve Y over S . If f has an isolated singularity at a closed point $x_0 \in |X|$, there is an invariant $\mu(X/Y, x_0)$ supported on x_0 , called the Milnor number. The Milnor formula [6, Théorème 2.4] proved by Deligne says that the Milnor number is related to the total dimension at x_0 of the vanishing cycles $R\Phi(\Lambda, f)$ of f for the constant sheaf Λ , i.e.,

$$(1.1.1) \quad (-1)^n \mu(X/Y, x_0) = -\dim_{\text{tot}} R\Phi_{\bar{x}_0}(\Lambda, f),$$

where $n = \dim X$ and $\dim_{\text{tot}} = \dim + \operatorname{Sw}$ denotes the total dimension. Later in [8], Deligne proposed a Milnor formula for any constructible sheaf $\mathcal{F}$ of Λ -modules on X , which is realized and proved by Saito in [13]. If $x_0 \in |X|$ is at most an isolated characteristic point of f with respect to the singular support of $\mathcal{F}$, then Saito's theorem [13, Theorem 5.9] says

$$(1.1.2) \quad (CC(\mathcal{F}), df)_{T^*X, x_0} = -\dim_{\text{tot}} R\Phi_{\bar{x}_0}(\mathcal{F}, f),$$

where $CC(\mathcal{F})$ is the characteristic cycle of $\mathcal{F}$. Now we propose the following question:

Question 1.2. Is there a Milnor type formula for non-isolated singular/characteristic points?

1.3. If f is a projective flat morphism and if f is smooth outside $f^{-1}(y)$ for a closed point y of the curve Y , then the conductor formula of Bloch (cf. [14, Theorem 2.2.3 and Corollary 2.2.4])

$$(1.3.1) \quad -a_y(Rf_*\Lambda) = (-1)^n(X, X)_{T^*X, X_y} = (-1)^n \deg c_{n, X_y}^X(\Omega_{X/Y}^1) \cap [X]$$

gives a partial answer to Question 1.2. We view (1.1.1), (1.1.2) and (1.3.1) in the form

$$(1.3.2) \quad \deg(\text{geometric class on singular locus}) = \deg(\text{cohomology class on singular locus}).$$

We expect the equality holds without taking degree, i.e.,

$$(1.3.3) \quad \text{cl}(\text{geometric class}) = \text{cohomology class},$$

where cl is the cycle class map. In the paper [18], the third author and Yigeng Zhao introduced a (cohomological) non-acyclicity class which is supported on the non-acyclicity locus. Let Y be a smooth scheme over k and $X \rightarrow Y$ a separated morphism over k . Let $Z \subseteq X$ be a closed subscheme and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ such that $X \setminus Z \rightarrow Y$ is universally locally acyclic relatively to $\mathcal{F}|_{X \setminus Z}$. Then the cohomological non-acyclicity class $\tilde{C}_{X/Y/k}^Z(\mathcal{F})$ is a class in $H_Z^0(X, \mathcal{K}_{X/Y/k})$, where $\mathcal{K}_{X/Y/k}$ sits in a distinguished triangle

$$(1.3.4) \quad \mathcal{K}_{X/Y} \rightarrow \mathcal{K}_{X/k} \rightarrow \mathcal{K}_{X/Y/k} \xrightarrow{+1}.$$

In this paper, we construct the geometric counterpart of the non-acyclicity class $\tilde{C}_{X/Y/k}^Z(\mathcal{F})$. More precisely, when $X \rightarrow Y$ is a morphism between smooth schemes of pure dimensions over k such that $X \rightarrow Y$ is $SS(\mathcal{F})$ -transversal outside Z , then we construct a class $cc_{X/Y/k}^Z(\mathcal{F}) \in \text{CH}_0(Z)$ (cf. (3.9.1)), called the geometric non-acyclicity class of $\mathcal{F}$. Explicitly, we put $r = \dim Y > 0$. Let $q : \mathbb{P}(T^*Y \times_Y X) \rightarrow X$ be the projective bundle, with the tautological line bundle $\mathcal{O}(-1)$ over it. Let $\bar{d}f$ be the section of $q^*T^*X \otimes \mathcal{O}(1)$ induced by $df : T^*Y \times_Y X \rightarrow T^*X$. Write $CC(\mathcal{F})_{\mathcal{O}(1)}$ for the conical cycle locally identified with $q^*CC(\mathcal{F})$ after trivializing $\mathcal{O}(1)$, cf. section 3.8. Define

$$(1.3.5) \quad cc_{X/Y/k}^Z(\mathcal{F}) := q_{Z*} \left[c_{r-1} \left(\frac{q^* f^* T^*(Y/k)}{\mathcal{O}(-1)} \right) \cap \bar{d}f^! CC(\mathcal{F})_{\mathcal{O}(1)} \right] \in \text{CH}_0(Z).$$

By $SS(\mathcal{F})$ -transversal condition of f outside Z , we may define the (geometric) relative characteristic class $cc_{X \setminus Z/Y}(\mathcal{F}) \in \text{CH}_{\dim Y}(X \setminus Z)$ which has a canonical extension $\bar{c}c_{X/Y}(\mathcal{F}) \in \text{CH}_{\dim Y}(X)$. For $\tau : Z \hookrightarrow X$, we have the fibration formula (3.12.1)

$$(1.3.6) \quad cc_{X/k}(\mathcal{F}) = c_{\dim Y}(f^* \Omega_{Y/k}^{1, \vee}) \cap \bar{c}c_{X/Y}(\mathcal{F}) + \tau_* cc_{X/Y/k}^Z(\mathcal{F}).$$

In general, we may replace the Spec k with general smooth connected scheme S , and we have the following properties, cf. equation (3.12.1), Propositions 3.22, 3.23, and 3.14, Theorem 3.15, Corollary 3.24:

Theorem 1.4. *Let k be a perfect field of positive characteristic and Λ a finite local ring whose residue characteristic is invertible in k . Consider morphisms $f : X \rightarrow Y$, $g : Y \rightarrow S$ of smooth separated schemes of finite type over k , all of pure dimension. Assume that S is connected of dimension s and that g is smooth of pure relative dimension $r > 0$. Put $h = g \circ f$, and let $\tau : Z \hookrightarrow X$ be a closed immersion. Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ such that h is $SS(\mathcal{F})$ -transversal and $f|_{X \setminus Z}$ is $SS(\mathcal{F}|_{X \setminus Z})$ -transversal. Then the geometric non-acyclicity class $cc_{X/Y/S}^Z(\mathcal{F}) \in \text{CH}_s(Z)$ has the following properties.*

(1) (Fibration formula) Let

$$(1.4.1) \quad cc_{X/S}(\mathcal{F}) = (-1)^s dh^! CC(\mathcal{F})$$

be the geometric characteristic class. There is a canonical class $\overline{cc}_{X/Y}(\mathcal{F}) \in \mathrm{CH}_{r+s}(X)$ whose restriction to $X \setminus Z$ is $cc_{(X \setminus Z)/Y}(\mathcal{F}|_{X \setminus Z})$. We have

$$(1.4.2) \quad cc_{X/S}(\mathcal{F}) = c_r((f^* \Omega_{Y/S}^1)^\vee) \cap \overline{cc}_{X/Y}(\mathcal{F}) + \tau_* cc_{X/Y/S}^Z(\mathcal{F}) \quad \text{in } \mathrm{CH}_s(X).$$

- (2) (Pullback) Let $\delta : S' \rightarrow S$ be a morphism, where S' is smooth and connected of dimension s' over k . Put $X' = X \times_S S'$, $Y' = Y \times_S S'$ and $Z' = Z \times_S S'$, and let $i_X : X' \rightarrow X$ and $f' : X' \rightarrow Y'$ be the induced morphisms. Assume that X' and Y' are smooth and of pure dimension, that $\dim X' - \dim X = s' - s$, and that i_X is properly $SS(\mathcal{F})$ -transversal. Then we have

$$(1.4.3) \quad i_X^! cc_{X/Y/S}^Z(\mathcal{F}) = cc_{X'/Y'/S'}^{Z'}(i_X^* \mathcal{F}) \quad \text{in } \mathrm{CH}_{s'}(Z').$$

- (3) (Proper pushforward) Let $p : X \rightarrow X'$ be a proper morphism over Y , where X' is smooth and of pure dimension over k , and let $f' : X' \rightarrow Y$ satisfy $f = f' \circ p$. Let $Z' \subseteq X'$ be closed with $Z \subseteq p^{-1}(Z')$. Assume that $\dim p_* SS(\mathcal{F}) \leq \dim X'$, where $p_* SS(\mathcal{F})$ is the image of $(dp)^{-1} SS(\mathcal{F})$ under $T^* X' \times_{X'} X \rightarrow T^* X'$. Then $g \circ f'$ is $SS(Rp_* \mathcal{F})$ -transversal and $f'|_{X' \setminus Z'}$ is $SS((Rp_* \mathcal{F})|_{X' \setminus Z'})$ -transversal. Moreover, we have

$$(1.4.4) \quad p_* cc_{X/Y/S}^Z(\mathcal{F}) = cc_{X'/Y/S}^{Z'}(Rp_* \mathcal{F}) \quad \text{in } \mathrm{CH}_s(Z').$$

- (4) (Milnor formula) Assume $S = \mathrm{Spec} k$, Y is a smooth curve and Z is a finite set of closed points, then we have

$$(1.4.5) \quad cc_{X/Y/k}^Z(\mathcal{F}) = - \sum_{x \in Z} \dim_{\mathrm{tot}} R\Phi_{\bar{x}}(\mathcal{F}, f)[x] \quad \text{in } \mathrm{CH}_0(Z),$$

where $R\Phi(\mathcal{F}, f)$ is the complex of vanishing cycles and $\dim_{\mathrm{tot}} = \dim + \mathrm{Sw}$ is the total dimension.

- (5) (Conductor formula) Assume $S = \mathrm{Spec} k$, Y is a smooth connected curve, and f is proper. If $T \subset Y$ is a finite set of closed points and $Z = f^{-1}(T)$, then we have

$$(1.4.6) \quad f_* cc_{X/Y/k}^Z(\mathcal{F}) = - \sum_{y \in T} a_y(Rf_* \mathcal{F})[y] \quad \text{in } \mathrm{CH}_0(T),$$

where $a_y(\mathcal{G}) = \mathrm{rank} \mathcal{G}_{\bar{\eta}} - \mathrm{rank} \mathcal{G}_{\bar{y}} + \mathrm{Sw}_y(\mathcal{G})$ is the Artin conductor of $\mathcal{G} \in D_{\mathrm{ctf}}(Y, \Lambda)$ at y , and η is the generic point of Y .

In view of the similarity of $cc_{X/Y/k}^Z(\mathcal{F})$ to the cohomological non-acyclicity class, it is natural to formulate the following conjecture:

Conjecture 1.5 (Conjecture 3.16). *We have*

$$(1.5.1) \quad \tilde{C}_{X/Y/k}^Z(\mathcal{F}) = \tilde{\mathrm{cl}}(cc_{X/Y/k}^Z(\mathcal{F})) \quad \text{in } H_Z^0(X, \mathcal{K}_{X/Y/k}),$$

where $\tilde{\mathrm{cl}} : \mathrm{CH}_0(Z) \xrightarrow{\mathrm{cl}} H_Z^0(X, \mathcal{K}_{X/k}) \xrightarrow{(1.3.4)} H_Z^0(X, \mathcal{K}_{X/Y/k})$ and cl is the cycle class map.

We hope (1.5.1) gives a formulation of Question 1.2 in some sense.

When $Z = X$, let $\rho : H^0(X, \mathcal{K}_{X/k}) \rightarrow H^0(X, \mathcal{K}_{X/Y/k})$ be the map induced by (1.3.4) and let $0_X : X \hookrightarrow T^*X$ be the zero section. The conjecture becomes

$$(1.5.2) \quad \rho(\mathrm{cl}(0_X^! CC(\mathcal{F}))) = \rho(C_{X/k}(\mathcal{F})),$$

which is the image of a conjectured formula by Saito [13, Conjecture 6.8(1) and Lemma 6.9(1)] under ρ . When $f = \mathrm{id}_X$, X is irreducible and $U := X \setminus Z \neq \emptyset$, the conjecture becomes

$$(1.5.3) \quad C_Z^0(\mathcal{F}) = \mathrm{cl}\left(0_X^! (CC(\mathcal{F}) - \mathrm{rank}(\mathcal{F}|_U) \cdot CC(\Lambda))\right) \quad \text{in } H_Z^0(X, \mathcal{K}_{X/k}),$$

where $C_Z^0(\mathcal{F})$ is the localized characteristic class of Abbes–Saito [1, Definition 5.2.1], rank $\mathcal{F}$ is the generic rank and the difference of characteristic cycles is supported on $T^*X \times_X Z$. In fact, by [18, p. 163, immediately after Theorem 5.1],

$$(1.5.4) \quad \rho_Z : H_Z^0(X, \mathcal{K}_{X/k}) \longrightarrow H_Z^0(X, \mathcal{K}_{X/X/k})$$

is an isomorphism in this case, and we have $C_Z^0(\mathcal{F}) = \tilde{C}_{X/X/k}^Z(\mathcal{F})$.

In Section 4, we prove Conjecture 1.5 in the quasi-affine case for morphisms to the affine line over an algebraically closed field. More precisely, we have the following theorem.

Theorem 1.6 (Theorem 4.2). *Let k be an algebraically closed field of characteristic $p > 0$ and Λ a finite local ring such that the characteristic of its residue field is invertible in k . Let X be a smooth quasi-affine scheme of finite type over k , purely of dimension d . Let $f : X \rightarrow \mathbb{A}_k^1 = \text{Spec } k[u]$ be a morphism and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Put $Z = (df)^{-1}SS(\mathcal{F})$, where*

$$(1.6.1) \quad df = f^*du : X \rightarrow T^*\mathbb{A}_k^1 \times_{\mathbb{A}_k^1} X \rightarrow T^*X$$

is the section defined by the differential of f . Then we have

$$(1.6.2) \quad \tilde{C}_{X/\mathbb{A}_k^1/k}^Z(\mathcal{F}) = \tilde{\text{cl}}(df^!CC(\mathcal{F})) \quad \text{in } H_Z^0(X, \mathcal{K}_{X/\mathbb{A}_k^1/k}).$$

The quasi-affineness of X allows us to choose a global function g such that the generic member of the family $f + ug$ has only isolated characteristic points (Proposition 4.13 and Lemma 4.15). After passing to the perfect closure of the generic point of $\mathbb{A}_k^1$, the desired equality follows from Saito’s Milnor formula (3.15.1) and the cohomological Milnor formula (3.3.4). We then descend this equality to the generic point and specialize to $u = 0$. For this, we prove that the supported cycle class map is compatible with specialization (Theorem 4.8). Then theorem 1.6 follows from the specialization formulas for refined Gysin pullbacks and non-acyclicity classes.

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Notation and Conventions.

- (1) Let S be a Noetherian scheme and Sch_S the category of separated schemes of finite type over S .
- (2) Let Λ be a Noetherian ring such that $m\Lambda = 0$ for some integer m invertible on S unless otherwise stated explicitly.
- (3) For any scheme $X \in \text{Sch}_S$, we denote by $D_{\text{ctf}}(X, \Lambda)$ the derived category of complexes of Λ -modules of finite Tor-dimension with constructible cohomology sheaves on X .
- (4) For any separated morphism $f : X \rightarrow Y$ in Sch_S , we use the following notations

$$(1.6.3) \quad \mathcal{K}_{X/Y} = Rf^!\Lambda, \quad D_{X/Y}(-) = R\mathcal{H}om(-, \mathcal{K}_{X/Y}).$$

- (5) For $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(Y, \Lambda)$ on S -schemes X and Y respectively, $\mathcal{F} \boxtimes_S^L \mathcal{G}$ denotes $\text{pr}_1^*\mathcal{F} \otimes^L \text{pr}_2^*\mathcal{G}$ on $X \times_S Y$.
- (6) To simplify our notation, we omit R or L denoting the derived functors, unless stated explicitly or for $R\mathcal{H}om$.

2. TRANSVERSALITY CONDITION

2.1. We recall the (cohomological) transversality condition introduced in [18, §2.1], which is a relative version of the transversality condition studied by Saito [13, Definition 8.5]. Consider the following Cartesian diagram in Sch_S :

$$(2.1.1) \quad \begin{array}{ccc} X & \xrightarrow{i} & Y \\ p \downarrow & \square & \downarrow f \\ W & \xrightarrow{\delta} & T. \end{array}$$

Let $\mathcal{F} \in D_{\text{ctf}}(Y, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(T, \Lambda)$. Let $c_{\delta, f, \mathcal{F}, \mathcal{G}}$ be the composition

$$(2.1.2) \quad \begin{aligned} c_{\delta, f, \mathcal{F}, \mathcal{G}} : i^* \mathcal{F} \otimes^L p^* \delta^! \mathcal{G} &\xrightarrow{id \otimes b.c} i^* \mathcal{F} \otimes^L i^! f^* \mathcal{G} \\ &\xrightarrow{\text{adj}} i^! i_!(i^* \mathcal{F} \otimes^L i^! f^* \mathcal{G}) \\ &\xrightarrow[\simeq]{\text{proj. formula}} i^! (\mathcal{F} \otimes^L i_! i^! f^* \mathcal{G}) \xrightarrow{\text{adj}} i^! (\mathcal{F} \otimes^L f^* \mathcal{G}). \end{aligned}$$

We put $c_{\delta, f, \mathcal{F}} := c_{\delta, f, \mathcal{F}, \Lambda} : i^* \mathcal{F} \otimes^L p^* \delta^! \Lambda \rightarrow i^! \mathcal{F}$. If $c_{\delta, f, \mathcal{F}}$ is an isomorphism, then we say that the morphism δ is $\mathcal{F}$ -transversal. If $c_{i, id, \mathcal{F}}$ is an isomorphism, then we say i is $\mathcal{F}$ -transversal.

By [18, §2.3], there is a functor $\delta^\Delta : D_{\text{ctf}}(Y, \Lambda) \rightarrow D_{\text{ctf}}(X, \Lambda)$ such that for any $\mathcal{F} \in D_{\text{ctf}}(Y, \Lambda)$, we have a distinguished triangle

$$(2.1.3) \quad i^* \mathcal{F} \otimes^L p^* \delta^! \Lambda \xrightarrow{c_{\delta, f, \mathcal{F}}} i^! \mathcal{F} \rightarrow \delta^\Delta \mathcal{F} \xrightarrow{+1}.$$

Then δ is $\mathcal{F}$ -transversal if and only if $\delta^\Delta(\mathcal{F}) = 0$ (cf. [18, Lemma 2.7]).

The following lemma gives an equivalence characterization between transversality condition and (universally) locally acyclicity condition.

Lemma 2.2. *Let $f : X \rightarrow S$ be a separated morphism of finite type between Noetherian schemes and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. The following conditions are equivalent:*

- (1) *The morphism f is locally acyclic relatively to $\mathcal{F}$.*
- (2) *The morphism f is universally locally acyclic relatively to $\mathcal{F}$.*
- (3) *For any $\mathcal{G} \in D_{\text{ctf}}(X, \Lambda)$, the canonical map*

$$(2.2.1) \quad D_{X/S}(\mathcal{G}) \boxtimes_S^L \mathcal{F} \rightarrow R\mathcal{H}om_{X \times_S X}(\text{pr}_1^* \mathcal{G}, \text{pr}_2^! \mathcal{F})$$

is an isomorphism in $D_{\text{ctf}}(X \times_S X, \Lambda)$, where $\text{pr}_1 : X \times_S X \rightarrow X$ and $\text{pr}_2 : X \times_S X \rightarrow X$ are the projections.

- (4) *The canonical map*

$$(2.2.2) \quad D_{X/S}(\mathcal{F}) \boxtimes_S^L \mathcal{F} \rightarrow R\mathcal{H}om_{X \times_S X}(\text{pr}_1^* \mathcal{F}, \text{pr}_2^! \mathcal{F})$$

is an isomorphism.

- (5) *For any Cartesian diagram*

$$(2.2.3) \quad \begin{array}{ccc} Y \times_S X & \xrightarrow{\text{pr}_2} & X \\ \text{pr}_1 \downarrow & \square & \downarrow f \\ Y & \xrightarrow{\delta} & S \end{array}$$

such that all schemes are Noetherian and δ is separated of finite type, the morphism δ is $\mathcal{F}$ -transversal.

- (6) For any Cartesian diagram (2.2.3) and any $\mathcal{G} \in D_{\text{ctf}}(S, \Lambda)$, the morphism $c_{\delta, f, \mathcal{F}, \mathcal{G}}$ is an isomorphism.
- (7) For any Cartesian diagram

$$(2.2.4) \quad \begin{array}{ccccc} Y \times_S X & \xrightarrow{\text{pr}_2} & X' & \longrightarrow & X \\ \text{pr}_1 \downarrow & \square & \downarrow f' & \square & \downarrow f \\ Y & \xrightarrow{\delta} & S' & \longrightarrow & S, \end{array}$$

such that all schemes are Noetherian and δ is separated of finite type, the morphism δ is $\mathcal{F}|_{X'}$ -transversal.

- (8) For any Cartesian diagram (2.2.4) and any $\mathcal{G} \in D_{\text{ctf}}(S', \Lambda)$, the morphism $c_{\delta, f', \mathcal{F}|_{X'}, \mathcal{G}}$ is an isomorphism.

When S is a scheme of finite type over a field k , then the equivalence between (2) and (7) follows from [18, Propositions 2.3(2) and 2.4]. In this case, we may require Y and S' smooth over k in (7).

Proof. By a result of Gabber [10, Corollary 6.6], (1) and (2) are equivalent. The equivalence between (2), (3), and (4) follows from [11, Proposition 2.5, Lemma 2.14, and Theorem 2.16]. By [18, Proposition 2.3(2)], (2) implies (6). It is clear that (6) implies (5).

Now we show (5) implies (1). By [11, Theorem 2.16], it suffices to show that for any $\mathcal{H} \in D_{\text{ctf}}(Y, \Lambda)$, the canonical morphism

$$(2.2.5) \quad D_{Y/S}(\mathcal{H}) \boxtimes_S^L \mathcal{F} \rightarrow R\mathcal{H}om(\text{pr}_1^* \mathcal{H}, \text{pr}_2^! \mathcal{F})$$

is an isomorphism. Using a standard dévissage argument [3, Exposé IX, Proposition 2.7], we may assume $\mathcal{H} = j_* \Lambda$ for $j : U \rightarrow Y$ étale with U affine. In this case, (2.2.5) is the following composition

$$(2.2.6) \quad j_* j^* \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F} \rightarrow (j \times \text{id})_* (j^* \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F}) \xrightarrow[\simeq]{(2.2.8)} (j \times \text{id})_* (j \times \text{id})^* \text{pr}_2^! \mathcal{F}.$$

In the following we show that

$$(2.2.7) \quad j_* j^* \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F} \rightarrow (j \times \text{id})_* (j^* \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F})$$

is an isomorphism. We write j as a composition of a proper morphism and an open immersion.

Since δ is $\mathcal{F}$ -transversal, we have an isomorphism

$$(2.2.8) \quad \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F} \xrightarrow{\simeq} \text{pr}_2^! \mathcal{F}.$$

By the projection formula and proper base change, for any proper morphism $g : Y' \rightarrow Y$ and $A \in D_{\text{ctf}}(Y', \Lambda)$, the following canonical morphism

$$(2.2.9) \quad g_* A \boxtimes_S^L \mathcal{F} \xrightarrow{\simeq} (g \times \text{id})_! (A \boxtimes_S^L \mathcal{F})$$

is an isomorphism. If $u : Y' \rightarrow Y$ is an open immersion with complementary closed immersion $\tau : Z = Y \setminus Y' \rightarrow Y$, we have a commutative diagram between distinguished triangles

$$(2.2.10) \quad \begin{array}{ccccccc} \tau_* \tau^! \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F} & \longrightarrow & \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F} & \longrightarrow & u_* u^* \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F} & \xrightarrow{+1} & \longrightarrow \\ \textcolor{red}{(2.2.9)} \downarrow \simeq & & \downarrow \simeq & & \downarrow & & \\ (\tau \times \text{id})_! (\tau^! \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F}) & \longrightarrow & \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F} & \longrightarrow & (u \times \text{id})_* (u^* \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F}) & \xrightarrow{+1} & \longrightarrow \\ \textcolor{red}{(2.2.8)} \downarrow \simeq & & \parallel & & \downarrow \simeq & & \\ (\tau \times \text{id})_! (\tau \times \text{id})^! (\mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F}) & \longrightarrow & \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F} & \longrightarrow & (u \times \text{id})_* (u \times \text{id})^* (\mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F}) & \xrightarrow{+1} & \longrightarrow. \end{array}$$

Thus

$$(2.2.11) \quad u_* u^* \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F} \rightarrow (u \times \text{id})_*(u^* \mathcal{K}_{Y/S} \boxtimes_S^L \mathcal{F})$$

is an isomorphism. Applying (2.2.11) and (2.2.9), we obtain that (2.2.7) is an isomorphism, and the conclusion follows.

We show (2) implies (8). By assumption, f' is also universally locally acyclic relatively to $\mathcal{F}|_{X'}$. Thus by (6), the morphism $c_{\delta, f', \mathcal{F}|_{X'}, \mathcal{G}}$ is an isomorphism.

Finally, it is clear that (8) implies (7) and (7) implies (5). $\square$

2.3. Now we recall the geometric transversality condition (cf. [4, 1.2] and [13, Definitions 3.3, 3.5, and 7.1]). Let X be a smooth scheme over a field k . Let C be a conical closed subset of T^*X , i.e., a closed subset which is stable under the action of the multiplicative group $\mathbb{G}_m$. We denote by $T_X^*X \subseteq T^*X$ the zero section of the cotangent bundle T^*X of X .

- (1) Let $h: W \rightarrow X$ be a morphism from a smooth scheme W over k . We say that h is C -transversal if the intersection $(C \times_X W) \cap dh^{-1}(T_W^*W)$ is contained in the zero-section $T_X^*X \times_X W \subseteq T^*X \times_X W$, where $dh: T^*X \times_X W \rightarrow T^*W$ is the canonical map.

If h is C -transversal, we define $h^\circ C$ to be the image of $C \times_X W$ under the map $dh: T^*X \times_X W \rightarrow T^*W$. By [13, Lemma 3.1], $h^\circ C$ is a closed conical subset of T^*W .

- (2) Assume that X and C are purely of dimension d and that W is purely of dimension m . We say that a C -transversal map $h: W \rightarrow X$ is properly C -transversal if every irreducible component of $C \times_X W$ is of dimension m .
- (3) We say that a morphism $f: X \rightarrow Y$ to a smooth scheme Y over k is C -transversal if the inverse image $df^{-1}(C)$ is contained in the zero-section $T_Y^*Y \times_Y X \subseteq T^*Y \times_Y X$, where $df: T^*Y \times_Y X \rightarrow T^*X$ is the canonical map.

If f is proper over the base $C \cap T_X^*X$ of C , we define $f_\circ C$ to be the image of $(df)^{-1}(C)$ under the projection $\text{pr}: T^*Y \times_Y X \rightarrow T^*Y$. In this case, $f_\circ C$ is also a closed conical subset of T^*Y .

The cohomological transversality condition and geometric transversality condition are related as follows. Let X be a smooth scheme of pure dimension d and Λ a finite local ring such that the characteristic ℓ of the residue field of Λ is invertible in k . Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. The singular support $SS(\mathcal{F})$ defined by Beilinson [4] is a d -dimensional conical closed subset of T^*X . We have the following properties:

- (1) $SS(\mathcal{F}) \cap T_X^*X = \text{supp}(\mathcal{F})$.
- (2) Let $f: X \rightarrow Y$ be a morphism to a smooth scheme Y over k . If f is $SS(\mathcal{F})$ -transversal, then f is universally locally acyclic relatively to $\mathcal{F}$.
- (3) Let $f: W \rightarrow X$ be a morphism from a smooth scheme W over k . If f is $SS(\mathcal{F})$ -transversal, then f is $\mathcal{F}$ -transversal.

Now assume k is a perfect field. By [13, Theorem 5.9 and Theorem 5.18], the characteristic cycle $CC(\mathcal{F})$ is the unique d -cycle $CC(\mathcal{F})$ supported on $SS(\mathcal{F})$ with $\mathbb{Z}$ -coefficients such that $CC(\mathcal{F})$ satisfies the Milnor formula (1.1.2) for $\mathcal{F}$.

3. GEOMETRIC NON-ACYCLICITY CLASSES

3.1. Let S be a Noetherian scheme and Λ a Noetherian ring such that $m\Lambda = 0$ for some integer m invertible on S . Consider the following Cartesian diagram in Sch_S

$$(3.1.1) \quad \begin{array}{ccc} X \times_S Y & \xrightarrow{\text{pr}_1} & X \\ \text{pr}_2 \downarrow & \square & \downarrow h \\ Y & \xrightarrow{g} & S, \end{array}$$

where pr_1 and pr_2 are the projections. For any $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and $\mathcal{G} \in D_{\text{ctf}}(Y, \Lambda)$, we have canonical morphisms

$$(3.1.2) \quad \mathcal{F} \boxtimes_S^L \mathcal{K}_{Y/S} = \text{pr}_1^* \mathcal{F} \otimes^L \text{pr}_2^* g^! \Lambda \xrightarrow{c_{g,h,\mathcal{F}}} \text{pr}_1^! \mathcal{F},$$

$$(3.1.3) \quad \mathcal{F} \boxtimes_S^L D_{Y/S}(\mathcal{G}) \rightarrow R\mathcal{H}om_{X \times_S Y}(\text{pr}_2^* \mathcal{G}, \text{pr}_1^! \mathcal{F}),$$

where (3.1.3) is adjoint to

$$(3.1.4) \quad \mathcal{F} \boxtimes_S^L (D_{Y/S}(\mathcal{G}) \otimes^L \mathcal{G}) \xrightarrow{id \boxtimes \text{ev}} \mathcal{F} \boxtimes_S^L \mathcal{K}_{Y/S} \xrightarrow{(3.1.2)} \text{pr}_1^! \mathcal{F}.$$

Note that (3.1.2) is a special case of (3.1.3) by taking $\mathcal{G} = \Lambda$. If moreover $X \rightarrow S$ is universally locally acyclic relatively to $\mathcal{F}$, then (3.1.3) is an isomorphism by (2.2.5). For a morphism $c = (c_1, c_2) : C \rightarrow X \times_S Y$, we have a canonical isomorphism by [5, Corollaire 3.1.12.2]

$$(3.1.5) \quad R\mathcal{H}om(c_2^* \mathcal{G}, c_1^! \mathcal{F}) \xrightarrow{\sim} c^! R\mathcal{H}om(\text{pr}_2^* \mathcal{G}, \text{pr}_1^! \mathcal{F}).$$

3.2. Consider a commutative diagram in Sch_S :

$$(3.2.1) \quad \begin{array}{ccccc} Z & \xhookrightarrow{\tau} & X & \xrightarrow{f} & Y \\ & & \searrow h & & \swarrow g \\ & & S & & \end{array}$$

where $\tau : Z \rightarrow X$ is a closed immersion and g is a smooth morphism. Let $i : X \times_Y X \rightarrow X \times_S X$ be the base change of the diagonal morphism $\delta : Y \rightarrow Y \times_S Y$:

$$(3.2.2) \quad \begin{array}{ccccc} X & \xlongequal{\quad} & X \\ \delta_1 \downarrow & \square & \downarrow \delta_0 \\ f \downarrow X \times_Y X & \xrightarrow{i} & X \times_S X \\ p \downarrow & \square & \downarrow f \times f \\ Y & \xrightarrow{\delta} & Y \times_S Y, \end{array}$$

where δ_0 and δ_1 are the diagonal morphisms. Put $\mathcal{K}_{X/Y/S} := \delta^\Delta \mathcal{K}_{X/S} \simeq \delta_1^* \delta^\Delta \delta_{0*} \mathcal{K}_{X/S}$. By (2.1.3), we have the following distinguished triangle (see also [18, (4.6)])

$$(3.2.3) \quad \mathcal{K}_{X/Y} \rightarrow \mathcal{K}_{X/S} \rightarrow \mathcal{K}_{X/Y/S} \xrightarrow{+1}.$$

Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ such that $X \setminus Z \rightarrow Y$ is universally locally acyclic relatively to $\mathcal{F}|_{X \setminus Z}$ and that $h : X \rightarrow S$ is universally locally acyclic relatively to $\mathcal{F}$. We put

$$(3.2.4) \quad \mathcal{H}_S = R\mathcal{H}om_{X \times_S X}(\text{pr}_2^* \mathcal{F}, \text{pr}_1^! \mathcal{F}), \quad \mathcal{T}_S = \mathcal{F} \boxtimes_S^L D_{X/S}(\mathcal{F}).$$

The relative cohomological characteristic class $C_{X/S}(\mathcal{F})$ is the composition (cf. [18, (3.49)])

$$(3.2.5) \quad \Lambda \xrightarrow{\text{id}} R\mathcal{H}om(\mathcal{F}, \mathcal{F}) \xrightarrow[\simeq]{(3.1.5)} \delta_0^! \mathcal{H}_S \xleftarrow[\simeq]{(3.1.3)} \delta_0^! \mathcal{T}_S \rightarrow \delta_0^* \mathcal{T}_S \xrightarrow{\text{ev}} \mathcal{K}_{X/S}.$$

By the assumption on $\mathcal{F}$, $\delta_1^* \delta^\Delta \mathcal{T}_S$ is supported on Z by [18, §4.3]. The non-acyclicity class $\tilde{C}_{X/Y/S}^Z(\mathcal{F})$ is the composition (cf. [18, Definition 4.2])

$$(3.2.6) \quad \Lambda \rightarrow \delta_0^! \mathcal{H}_S \xleftarrow{\simeq} \delta_0^! \mathcal{T}_S \simeq \delta_1^! i^! \mathcal{T}_S \rightarrow \delta_1^* i^! \mathcal{T}_S \rightarrow \delta_1^* \delta^\Delta \mathcal{T}_S \xleftarrow{\simeq} \tau_* \tau^! \delta_1^* \delta^\Delta \mathcal{T}_S \rightarrow \tau_* \tau^! \mathcal{K}_{X/Y/S}.$$

If the following condition holds:

$$(3.2.7) \quad H^0(Z, \mathcal{K}_{Z/Y}) = 0 \text{ and } H^1(Z, \mathcal{K}_{Z/Y}) = 0$$

then the map

$$(3.2.8) \quad H_Z^0(X, \mathcal{K}_{X/S}) \xrightarrow{(3.2.3)} H_Z^0(X, \mathcal{K}_{X/Y/S})$$

is an isomorphism. In this case, the class $\tilde{C}_{X/Y/S}^Z(\mathcal{F}) \in H_Z^0(X, \mathcal{K}_{X/Y/S})$ defines an element of $H_Z^0(X, \mathcal{K}_{X/S})$, which is denoted by $C_{X/Y/S}^Z(\mathcal{F})$. Now we summarize the functorial properties for the non-acyclicity classes.

Proposition 3.3 ([18, Theorem 1.5, Proposition 1.6, Theorem 1.7 Theorem 1.8 and Proposition 4.7]). *Let us denote the diagram (3.2.1) by $\Delta = \Delta_{X/Y/S}^Z$ and $\tilde{C}_{X/Y/S}^Z(\mathcal{F})$, $C_{X/Y/S}^Z(\mathcal{F})$ by $\tilde{C}_\Delta(\mathcal{F})$, $C_\Delta(\mathcal{F})$, respectively. Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Assume that $Y \rightarrow S$ is smooth of pure relative dimension r , $X \setminus Z \rightarrow Y$ is universally locally acyclic relatively to $\mathcal{F}|_{X \setminus Z}$ and that $X \rightarrow S$ is universally locally acyclic relatively to $\mathcal{F}$.*

(1) (Fibration formula) *If $H^0(Z, \mathcal{K}_{Z/Y}) = H^1(Z, \mathcal{K}_{Z/Y}) = 0$, then we have*

$$(3.3.1) \quad C_{X/S}(\mathcal{F}) = c_r(f^* \Omega_{Y/S}^{1, \vee}) \cap C_{X/Y}(\mathcal{F}) + C_\Delta(\mathcal{F}) \quad \text{in} \quad H^0(X, \mathcal{K}_{X/S}).$$

(2) (Pullback) *Let $b : S' \rightarrow S$ be a morphism of Noetherian schemes. Let $\Delta' = \Delta_{X'/Y'/S'}^{Z'}$ be the base change of $\Delta = \Delta_{X/Y/S}^Z$ by $b : S' \rightarrow S$. Let $b_X : X' = X \times_S S' \rightarrow X$ be the base change of b by $X \rightarrow S$. Then we have*

$$(3.3.2) \quad b_X^* \tilde{C}_\Delta(\mathcal{F}) = \tilde{C}_{\Delta'}(b_X^* \mathcal{F}) \quad \text{in} \quad H_{Z'}^0(X', \mathcal{K}_{X'/Y'/S'}),$$

where $b_X^* : H_Z^0(X, \mathcal{K}_{X/Y/S}) \rightarrow H_{Z'}^0(X', \mathcal{K}_{X'/Y'/S'})$ is the induced pullback morphism.

(3) (Proper pushforward) *Consider a diagram $\Delta' = \Delta_{X'/Y'/S'}^{Z'}$. Let $s : X \rightarrow X'$ be a proper morphism over Y such that $Z \subseteq s^{-1}(Z')$. Then we have*

$$(3.3.3) \quad s_*(\tilde{C}_\Delta(\mathcal{F})) = \tilde{C}_{\Delta'}(Rs_* \mathcal{F}) \quad \text{in} \quad H_{Z'}^0(X', \mathcal{K}_{X'/Y'/S'}),$$

where $s_* : H_Z^0(X, \mathcal{K}_{X/Y/S}) \rightarrow H_{Z'}^0(X', \mathcal{K}_{X'/Y'/S'})$ is the induced pushforward morphism.

(4) (Cohomological Milnor formula) *Assume $S = \text{Spec } k$ for a perfect field k of characteristic $p > 0$, Y is a smooth curve over k and Λ is a finite local ring such that the characteristic of the residue field is invertible in k . If $Z = \{x\}$, then we have*

$$(3.3.4) \quad C_\Delta(\mathcal{F}) = -\text{dimtot } R\Phi_{\bar{x}}(\mathcal{F}, f) \quad \text{in} \quad \Lambda = H_x^0(X, \mathcal{K}_{X/k}),$$

where $R\Phi(\mathcal{F}, f)$ is the complex of vanishing cycles and $\text{dimtot} = \text{dim} + \text{Sw}$ is the total dimension.

- (5) (*Cohomological conductor formula*) Assume $S = \operatorname{Spec} k$ for a perfect field k of characteristic $p > 0$ and Λ is a finite local ring such that the characteristic of the residue field is invertible in k . If Y is a smooth connected curve over k , f is proper and $Z = f^{-1}(y)$ for a closed point $y \in |Y|$, then we have

$$(3.3.5) \quad f_* \tilde{C}_\Delta(\mathcal{F}) = -a_y(Rf_* \mathcal{F}) \quad \text{in} \quad \Lambda = H_y^0(Y, \mathcal{K}_{Y/k}),$$

where $a_y(\mathcal{G}) = \operatorname{rank} \mathcal{G}|_{\bar{\eta}} - \operatorname{rank} \mathcal{G}_{\bar{y}} + \operatorname{Sw}_y \mathcal{G}$ is the Artin conductor of an object $\mathcal{G} \in D_{\text{ctf}}(Y, \Lambda)$ at y and η is the generic point of Y .

- (6) (*Specialization*) Assume that S is a henselian trait with generic point η , and let s be a geometric point above its closed point. Put $\Delta_t = \Delta_{X_t/Y_t/t}^{Z_t}$ and $\mathcal{F}_t = \mathcal{F}|_{X_t}$ for $t \in \{\eta, s\}$. For this item, the hypothesis on f may be weakened to universal local acyclicity of f_t relatively to $\mathcal{F}_t$ outside Z_t . Then we have

$$(3.3.6) \quad \operatorname{sp}^{\text{ét}}(\tilde{C}_{\Delta_\eta}(\mathcal{F}_\eta)) = \tilde{C}_{\Delta_s}(\mathcal{F}_s) \quad \text{in} \quad H_{Z_s}^0(X_s, \mathcal{K}_{X_s/Y_s/s}),$$

where

$$(3.3.7) \quad \operatorname{sp}^{\text{ét}} : H_{Z_\eta}^0(X_\eta, \mathcal{K}_{X_\eta/Y_\eta/\eta}) \longrightarrow H_{Z_s}^0(X_s, \mathcal{K}_{X_s/Y_s/s})$$

is the specialization map.

3.4. Let X be a smooth connected curve over k . Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ and $Z \subseteq X$ be a finite set of closed points such that the cohomology sheaves of $\mathcal{F}|_{X \setminus Z}$ are locally constant. By the cohomological Milnor formula (3.3.4), we have the following (motivic) expression for the Artin conductor of $\mathcal{F}$ at $x \in Z$

$$(3.4.1) \quad a_x(\mathcal{F}) = \dim_{\text{tot}} R\Phi_{\bar{x}}(\mathcal{F}, \text{id}) = -C_{U/U/k}^{\{x\}}(\mathcal{F}|_U),$$

where U is any open subscheme of X such that $U \cap Z = \{x\}$. By (3.3.1), we get the following cohomological Grothendieck-Ogg-Shafarevich formula (cf. [18, Corollary 6.4]):

$$(3.4.2) \quad C_{X/k}(\mathcal{F}) = \operatorname{rank} \mathcal{F} \cdot c_1(\Omega_{X/k}^{1,\vee}) - \sum_{x \in Z} a_x(\mathcal{F}) \cdot [x] \quad \text{in} \quad H^0(X, \mathcal{K}_{X/k}).$$

3.5. Now we start to construct and prove a geometric counterpart of Proposition 3.3. Let k be a perfect field of characteristic $p > 0$ and Λ be a finite local ring whose residue field is of characteristic $\ell \neq p$. Let Sm_k be the category of smooth schemes over k . Let S be a smooth connected scheme of dimension s over k . Let $f : X \rightarrow S$ be a morphism in Sm_k . Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ such that f is $SS(\mathcal{F})$ -transversal. Consider the following morphisms

$$(3.5.1) \quad X \xrightarrow{0} T^*S \times_S X \xrightarrow{df} T^*X,$$

where 0 stands for the zero section. By assumption, $df^{-1}(SS(\mathcal{F}))$ is contained in $0(X)$. We define the relative characteristic class of $\mathcal{F}$ to be the following s -cycle class on X :

$$(3.5.2) \quad cc_{X/S}(\mathcal{F}) := (-1)^s \cdot (df)^!(CC(\mathcal{F})) \quad \text{in} \quad \operatorname{CH}_s(X),$$

where $(df)^!$ is the refined Gysin pullback. When f is a smooth morphism, then we have a Cartesian diagram

$$(3.5.3) \quad \begin{array}{ccc} T^*S \times_S X & \xrightarrow{df} & T^*X \\ \downarrow & \square & \downarrow \\ X & \xrightarrow{0_{X/S}} & T^*(X/S). \end{array}$$

In this case, we have $cc_{X/S}(\mathcal{F}) = (-1)^s \cdot 0_{X/S}^!(CC(\mathcal{F}))$ (cf. [17, Definition 2.11]). If f is a smooth morphism of relative dimension r and if $\mathcal{F}$ is locally constant, then we have

$$(3.5.4) \quad cc_{X/S}(\mathcal{F}) = (-1)^s \cdot 0_{X/S}^!((-1)^{\dim X} \cdot \text{rank} \mathcal{F} \cdot [T_X^* X]) = \text{rank} \mathcal{F} \cdot c_r(\Omega_{X/S}^{1,\vee}) \cap [X].$$

We have the following conjecture, cf. [17, Conjecture 3.8]:

Conjecture 3.6. *Let S be a smooth connected scheme of dimension s over k . Let $f : X \rightarrow S$ be a morphism in Sm_k . Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ such that f is $SS(\mathcal{F})$ -transversal. Then we have*

$$(3.6.1) \quad \text{cl}(cc_{X/S}(\mathcal{F})) = C_{X/S}(\mathcal{F}) \quad \text{in} \quad H^0(X, \mathcal{K}_{X/S}),$$

where $\text{cl} : \text{CH}_s(X) \rightarrow H^0(X, \mathcal{K}_{X/S})$ is the cycle class map.

When $S = \text{Spec} k$, then it is Saito's conjecture [13, Conjecture 6.8.1], which is proved under quasi-projective assumption in [18, Theorem 1.2]. When $f : X \rightarrow S$ is a smooth morphism, then (3.6.1) is true for a locally constant constructible (flat) sheaf $\mathcal{F}$ of Λ -modules. Indeed, this follows from (3.5.4), [18, Lemma 3.1] and (3.3.1).

Remark 3.7. By [9, Chapter 17], $\text{CH}_s(X) \cong \text{CH}^0(X \rightarrow S)$ where the latter bivariant Chow group can be regarded as a relative version of $\text{CH}_0(X)$. However, in general we cannot expect $cc_{X/S}(\mathcal{F})$ to lift canonically to a relative cycle.

3.8. Consider a commutative diagram

$$(3.8.1) \quad \begin{array}{ccccc} Z & \xhookrightarrow{\tau} & X & \xrightarrow{f} & Y \\ & & \searrow h & & \swarrow g \\ & & S & & \end{array}$$

where X, Y, S belong to Sm_k and $\tau : Z \hookrightarrow X$ is a closed immersion. Assume that X, Y, S are of pure dimension, that S is connected of dimension s , and that g is smooth of relative dimension $r > 0$. Put $n = \dim X$. Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ such that h is $SS(\mathcal{F})$ -transversal and $f|_{X \setminus Z}$ is $SS(\mathcal{F}|_{X \setminus Z})$ -transversal. Let

$$(3.8.2) \quad q : \mathbb{P}(T^*Y \times_Y X) \longrightarrow X.$$

be the projective bundle induced by $T^*Y \times_Y X$ over X . The tautological subbundle and the differential give

$$(3.8.3) \quad \mathcal{O}(-1) \hookrightarrow q^*(T^*Y \times_Y X) \xrightarrow{q^*df} q^*T^*X.$$

Under $\text{Hom}(\mathcal{O}(-1), q^*T^*X) = q^*T^*X \otimes \mathcal{O}(1)$, this homomorphism defines a section of the vector bundle $q^*T^*X \otimes \mathcal{O}(1)$:

$$(3.8.4) \quad \overline{df} : \mathbb{P}(T^*Y \times_Y X) \longrightarrow q^*T^*X \otimes \mathcal{O}(1).$$

At a geometric point (x, λ) , with $\lambda \subset T_{f(x)}^*Y$ a line, its value is the map $df_x|_\lambda : \lambda \rightarrow T_x^*X$. Thus $\overline{df}$ is defined even when $df_x(\lambda) = 0$. It is a section of a vector bundle.

For a conical closed subscheme $D \subset T^*X$, we define its twist. On an open $V \subset \mathbb{P}(T^*Y \times_Y X)$ with local basis e of $\mathcal{O}(1)$, transport $D \times_X V$ by

$$(3.8.5) \quad q^*T^*X|_V \xrightarrow{\sim} (q^*T^*X \otimes \mathcal{O}(1))|_V, \quad \xi \mapsto \xi \otimes e.$$

Changing e multiplies the vector coordinate by a unit. Since the ideal of D is homogeneous, these closed subschemes agree on overlaps and define $D_{\mathcal{O}(1)} \subset q^*T^*X \otimes \mathcal{O}(1)$. Equivalently, if

$D = \text{Spec}_X(\bigoplus_{a \geq 0} \mathcal{A}_a)$ with its graded coordinate algebra, then

$$(3.8.6) \quad D_{\mathcal{O}(1)} = \text{Spec}_{\mathbb{P}(T^*Y \times_Y X)} \left(\bigoplus_{a \geq 0} q^* \mathcal{A}_a \otimes \mathcal{O}(-a) \right).$$

For $CC(\mathcal{F}) = \sum_a m_a [C_a]$, set

$$(3.8.7) \quad CC(\mathcal{F})_{\mathcal{O}(1)} := \sum_a m_a [(C_a)_{\mathcal{O}(1)}].$$

Each $(C_a)_{\mathcal{O}(1)}$ is integral of dimension $n + r + s - 1$. We apply the same twist separately to $SS(\mathcal{F})$.

Notice that we have the following Cartesian diagram

$$(3.8.8) \quad \begin{array}{ccc} \mathbb{P}((df)^{-1}SS(\mathcal{F})) & \longrightarrow & SS(\mathcal{F})_{\mathcal{O}(1)} \\ \downarrow & & \downarrow \\ \mathbb{P}(T^*Y \times_Y X) & \xrightarrow{\overline{df}} & q^*T^*X \otimes \mathcal{O}(1) \end{array}$$

which can be checked on open subschemes. Moreover, the section $\overline{df}$ is a regular embedding of codimension n , so

$$(3.8.9) \quad \overline{df}^! CC(\mathcal{F})_{\mathcal{O}(1)} \in \text{CH}_{r+s-1}(\mathbb{P}((df)^{-1}SS(\mathcal{F}))).$$

The hypotheses imply

$$(3.8.10) \quad (df)^{-1}SS(\mathcal{F}) \subset 0(X) \cup (T^*Y \times_Y Z), \quad (df)^{-1}SS(\mathcal{F}) \cap (T^*S \times_S X) \subset 0(X).$$

Consequently, on $\mathbb{P}((df)^{-1}SS(\mathcal{F}))$, the tautological line bundle $\mathcal{O}(-1)$ is a subbundle of $q^*f^*T^*(Y/S)$, by the short exact sequence of bundles

$$(3.8.11) \quad 0 \rightarrow T^*S \times_S X \rightarrow T^*Y \times_Y X \rightarrow f^*T^*(Y/S) \rightarrow 0.$$

The reduced closed subscheme of the support $\mathbb{P}((df)^{-1}SS(\mathcal{F}))$ maps properly to Z_{red} and hence to Z . For this pushforward, denoted by q_{Z*} , we use the canonical invariance of Chow groups under reduction. The refined intersection retains its scheme structure and intersection multiplicities.

Definition 3.9. We define the geometric non-acyclicity class of $\mathcal{F}$ by

$$(3.9.1) \quad cc_{X/Y/S}^Z(\mathcal{F}) := (-1)^s q_{Z*} \left[c_{r-1} \left(\frac{q^*f^*T^*(Y/S)}{\mathcal{O}(-1)} \right) \cap \overline{df}^! CC(\mathcal{F})_{\mathcal{O}(1)} \right] \in \text{CH}_s(Z).$$

Here, the quotient bundle is taken on $\mathbb{P}((df)^{-1}SS(\mathcal{F}))$.

Proposition 3.10. Consider the previous assumptions.

- (1) If f is $SS(\mathcal{F})$ -transversal on X , then $cc_{X/Y/S}^Z(\mathcal{F}) = 0$ for any Z ;
- (2) Let $Z \subset Z_1 \subset X$ be closed subschemes, with inclusion $i : Z \hookrightarrow Z_1$. Then

$$(3.10.1) \quad i_* cc_{X/Y/S}^Z(\mathcal{F}) = cc_{X/Y/S}^{Z_1}(\mathcal{F});$$

- (3) Let

$$(3.10.2) \quad \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \xrightarrow{+1}$$

be a distinguished triangle in $D_{\text{ctf}}(X, \Lambda)$ with $\mathcal{F}'$, $\mathcal{F}''$ satisfy the transversality condition in the beginning of section 3.8. Then we have

$$(3.10.3) \quad cc_{X/Y/S}^Z(\mathcal{F}) = cc_{X/Y/S}^Z(\mathcal{F}') + cc_{X/Y/S}^Z(\mathcal{F}'').$$

Proof. The parts (1) and (2) are direct from definition. For (3), we use [13, Lemma 5.13]. □

3.11. For a vector bundle V of rank d on X , put

$$(3.11.1) \quad c_t(V) = t^d + c_1(V)t^{d-1} + \cdots + c_d(V) \in \mathrm{CH}^*(X)[t].$$

We regard t as a formal variable of codimension one. Recall that

$$(3.11.2) \quad \frac{1}{c_t(V)} = t^{-d} \sum_{a \geq 0} s_a(V) t^{-a},$$

where $s(V) = c(V)^{-1}$ is the total Segre class.

Let $n = \dim X$, and write

$$(3.11.3) \quad 0_X : X \longrightarrow T^*X, \quad q_X : \mathbb{P}(T^*X) \longrightarrow X$$

for the zero section and the projection. Put $\xi_X = c_1(\mathcal{O}_{\mathbb{P}(T^*X)}(1))$. Write

$$(3.11.4) \quad CC(\mathcal{F}) = 0_{X*}\Gamma + \sum_a m_a[C_a],$$

where Γ is an n -cycle on X and no C_a is contained in the zero section. Since each C_a is conical, $\mathbb{P}(C_a)$ is a closed integral subscheme of $\mathbb{P}(T^*X)$ of dimension $n-1$. By the projective bundle formula, there are classes

$$(3.11.5) \quad \beta_j = q_{X*} \left[c_{n-1-j} \left(\frac{q_X^* T^* X}{\mathcal{O}(-1)} \right) \cap \sum_a m_a [\mathbb{P}(C_a)] \right], \quad 0 \leq j \leq n-1$$

such that

$$(3.11.6) \quad \sum_a m_a [\mathbb{P}(C_a)] = \sum_{j=0}^{n-1} \xi_X^j \cap q_X^* \beta_j \quad \text{in } \mathrm{CH}_{n-1}(\mathbb{P}(T^*X)).$$

Define $b_j \in \mathrm{CH}_j(X)$ by

$$(3.11.7) \quad \sum_{j=0}^n b_j t^j = c_t(T^*X) \cap \Gamma + \sum_{j=0}^{n-1} \beta_j t^j,$$

where

$$(3.11.8) \quad c_t(T^*X) \cap \Gamma = \sum_{i=0}^n (c_i(T^*X) \cap \Gamma) t^{n-i}.$$

In particular, $b_n = \Gamma$. The classes b_j are canonically determined by $CC(\mathcal{F})$. Indeed, by the projective bundle relation

$$(3.11.9) \quad \xi_X^n + c_1(T^*X)\xi_X^{n-1} + \cdots + c_n(T^*X) = 0$$

we have

$$(3.11.10) \quad \sum_{j=0}^n \xi_X^j \cap q_X^* b_j = \sum_a m_a [\mathbb{P}(C_a)].$$

Put $m = \dim Y = r + s$. We define

$$(3.11.11) \quad \overline{cc}_{X/Y}(\mathcal{F}) := (-1)^m \sum_{j \geq m} s_{j-m}(T^*Y \times_Y X) \cap b_j \in \mathrm{CH}_m(X).$$

Equivalently, $\overline{cc}_{X/Y}(\mathcal{F})$ is the constant coefficient of the Laurent expansion at $t = \infty$ of

$$(3.11.12) \quad (-1)^m \cdot c_t(T^*Y \times_Y X)^{-1} \cdot \sum_{j=0}^n b_j t^j.$$

Proposition 3.12 (fibration formula). *Consider the previous assumptions, then the restriction of $\overline{cc}_{X/Y}(\mathcal{F})$ to $X \setminus Z$ is $cc_{(X \setminus Z)/Y}(\mathcal{F}|_{X \setminus Z})$. Moreover,*

$$(3.12.1) \quad cc_{X/S}(\mathcal{F}) = c_r((f^*\Omega_{Y/S}^1)^\vee) \cap \overline{cc}_{X/Y}(\mathcal{F}) + \tau_*cc_{X/Y/S}^Z(\mathcal{F}) \quad \text{in } \mathrm{CH}_s(X).$$

In particular, if $\dim Z < r + s$, the restriction

$$(3.12.2) \quad \mathrm{CH}_{r+s}(X) \longrightarrow \mathrm{CH}_{r+s}(X \setminus Z)$$

is an isomorphism, and $\overline{cc}_{X/Y}(\mathcal{F})$ is the unique extension of $cc_{(X \setminus Z)/Y}(\mathcal{F}|_{X \setminus Z})$. If $Z = \emptyset$, then

$$(3.12.3) \quad cc_{X/S}(\mathcal{F}) = c_r((f^*\Omega_{Y/S}^1)^\vee) \cap cc_{X/Y}(\mathcal{F}).$$

Proof. Since $h : X \rightarrow S$ is $SS(\mathcal{F})$ -transversal, the inverse image of $SS(\mathcal{F})$ under

$$(3.12.4) \quad dh : T^*S \times_S X \longrightarrow T^*X$$

is contained in the zero section, hence every projectivized nonzero component of $CC(\mathcal{F})$ is disjoint from $\mathbb{P}(T^*S \times_S X)$. Hence (3.11.10) restricts to

$$(3.12.5) \quad \sum_j \xi^j \cap q^*b_j = 0$$

on $\mathbb{P}(T^*S \times_S X)$. By the projective bundle formula, the polynomial $\sum_j b_j t^j$ is divisible by $c_t(T^*S \times_S X)$. Write

$$(3.12.6) \quad \sum_{j=0}^n b_j t^j = c_t(T^*S \times_S X) \sum_{j \geq 0} a_j t^j, \quad a_j \in \mathrm{CH}_{s+j}(X).$$

The constant coefficient is the refined intersection with dh :

$$(3.12.7) \quad a_0 = (dh)^! CC(\mathcal{F}), \quad cc_{X/S}(\mathcal{F}) = (-1)^s a_0.$$

Since $g : Y \rightarrow S$ is smooth of relative dimension r , there is an exact sequence of vector bundles on X

$$(3.12.8) \quad 0 \longrightarrow T^*S \times_S X \longrightarrow T^*Y \times_Y X \longrightarrow f^*T^*(Y/S) \longrightarrow 0.$$

Hence

$$(3.12.9) \quad c_t(T^*Y \times_Y X) = c_t(T^*S \times_S X) c_t(f^*T^*(Y/S)).$$

It follows from (3.11.11) that

$$(3.12.10) \quad \overline{cc}_{X/Y}(\mathcal{F}) = (-1)^{r+s} \sum_{j \geq r} s_{j-r}(f^*T^*(Y/S)) \cap a_j.$$

The projection $T^*Y \times_Y X \longrightarrow f^*T^*(Y/S)$ has kernel $T^*S \times_S X$, and

$$(3.12.11) \quad (df)^{-1}SS(\mathcal{F}) \cap (T^*S \times_S X) \subset 0(X)$$

by the $SS(\mathcal{F})$ -transversality of h , hence it is finite on $(df)^{-1}SS(\mathcal{F})$. Let

$$(3.12.12) \quad q_S : \mathbb{P}(f^*T^*(Y/S)) \longrightarrow X$$

be the projective bundle projection, and put $\xi = c_1(\mathcal{O}_{\mathbb{P}(f^*T^*(Y/S))}(1))$. The preceding finite projection induces a finite morphism

$$(3.12.13) \quad \rho : \mathbb{P}((df)^{-1}SS(\mathcal{F})) \longrightarrow \mathbb{P}(f^*T^*(Y/S))$$

which identifies the tautological line bundles; hence we may slightly abuse the tautological bundle notation here. By the construction of the classes b_j we have

$$(3.12.14) \quad \rho_*(\overline{df}^! CC(\mathcal{F})_{\mathcal{O}(1)}) = \sum_{j \geq 0} \xi^j \cap q_S^* a_j.$$

On $\mathbb{P}(f^*T^*(Y/S))$ we have the canonical quotient bundle $q_S^*f^*T^*(Y/S)/\mathcal{O}(-1)$, projective bundle formula gives

(3.12.15)

$$(q_S)_* \left[c_{r-1} (q_S^*f^*T^*(Y/S) - \mathcal{O}(-1)) \cap \xi^j \right] = \begin{cases} 1, & j = 0, \\ 0, & 0 < j < r, \\ -c_r(f^*T^*(Y/S)) s_{j-r}(f^*T^*(Y/S)), & j \geq r. \end{cases}$$

Applying (3.12.15) and the definition of $cc_{X/Y/S}^Z(\mathcal{F})$, we obtain

$$(3.12.16) \quad \tau_* cc_{X/Y/S}^Z(\mathcal{F}) = (-1)^s \left[a_0 - c_r(f^*T^*(Y/S)) \cap \sum_{j \geq r} s_{j-r}(f^*T^*(Y/S)) \cap a_j \right].$$

By (3.12.10), the second term on the right is $c_r((f^*T^*(Y/S))^\vee) \cap \overline{cc}_{X/Y}(\mathcal{F})$. Together with $cc_{X/S}(\mathcal{F}) = (-1)^s a_0$, this proves (3.12.1).

It remains to prove the restriction assertion. Put $U = X \setminus Z$. On U , the morphism $f|_U : U \rightarrow Y$ is $SS(\mathcal{F}|_U)$ -transversal. Restricting (3.11.7) to U , the zero-section excess formula gives

$$(3.12.17) \quad \sum_j (b_j|_U) t^j = c_t(T^*Y \times_Y U) \sum_{j \geq 0} d_j t^j, \quad d_0 = (df)^\dagger CC(\mathcal{F}|_U).$$

Consequently,

$$(3.12.18) \quad \overline{cc}_{X/Y}(\mathcal{F})|_U = (-1)^m d_0 = cc_{U/Y}(\mathcal{F}|_U).$$

This proves the restriction statement.

Finally, if $\dim Z < r + s$, then $\mathrm{CH}_{r+s}(Z) = 0$. Hence we have an isomorphism

$$(3.12.19) \quad \mathrm{CH}_{r+s}(X) \xrightarrow{\sim} \mathrm{CH}_{r+s}(X \setminus Z),$$

by the localization sequence, and $\overline{cc}_{X/Y}(\mathcal{F})$ is the unique extension of $cc_{(X \setminus Z)/Y}(\mathcal{F}|_{X \setminus Z})$. Equation (3.12.3) is the special case $Z = \emptyset$. $\square$

Remark 3.13. In fact, the construction of $\overline{cc}_{X/Y}(\mathcal{F})$ can be regarded as a generalized definition of the characteristic class, weakening the transversality condition.

Proposition 3.14. *Let $S = \mathrm{Spec} k$, $Y = \mathbb{A}_k^1$ with coordinate t , and let Z contain the characteristic locus of f . Writing $df : X \times_{\mathbb{A}_k^1} T^*\mathbb{A}_k^1 \rightarrow T^*X$ for the section defined by f^*dt , one has*

$$(3.14.1) \quad cc_{X/\mathbb{A}_k^1/k}^Z(\mathcal{F}) = (df)^\dagger CC(\mathcal{F}) \quad \text{in } \mathrm{CH}_0(Z).$$

Proof. The section defined by dt identifies $\mathbb{P}(T^*\mathbb{A}_k^1 \times_{\mathbb{A}_k^1} X)$ with X , trivializes $\mathcal{O}(-1)$, and identifies $\overline{df}$ with df . In (3.9.1), $r = 1$, $s = 0$ and the Chern class involved is $c_0 = 1$. $\square$

Theorem 3.15 (Saito's isolated Milnor formula). *Assume $S = \mathrm{Spec} k$ and Y is a smooth curve. Let Z be a finite set of closed points of X , and assume f is $SS(\mathcal{F})$ -transversal outside Z . Then*

$$(3.15.1) \quad cc_{X/Y/k}^Z(\mathcal{F}) = - \sum_{x \in Z} \dim \mathrm{tot} R\Phi_{\bar{x}}(\mathcal{F}, f)[x] \quad \text{in } \mathrm{CH}_0(Z).$$

Here total dimension is computed after derived reduction of Λ to its residue field, with alternating signs for complexes.

Proof. Near $f(x)$ choose an étale coordinate on Y , we may assume $Y = \mathbb{A}_k^1$. The consequence follows from Saito's Milnor formula [13, Theorem 5.9]. $\square$

We expect the following Milnor type formula for non-isolated singular/characteristic points holds.

Conjecture 3.16. *Let S be a smooth connected k -scheme of dimension s . Consider the commutative diagram (3.8.1). Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ such that $X \setminus Z \rightarrow Y$ is $SS(\mathcal{F}|_{X \setminus Z})$ -transversal and that $X \rightarrow S$ is $SS(\mathcal{F})$ -transversal. Then we have an equality*

$$(3.16.1) \quad \tilde{C}_{X/Y/S}^Z(\mathcal{F}) = \tilde{\text{cl}}(cc_{X/Y/S}^Z(\mathcal{F})) \quad \text{in} \quad H_Z^0(X, \mathcal{K}_{X/Y/S}),$$

where $\tilde{\text{cl}}$ is the composition $\text{CH}_s(Z) \xrightarrow{\text{cl}} H_Z^0(X, \mathcal{K}_{X/S}) \xrightarrow{(3.2.3)} H_Z^0(X, \mathcal{K}_{X/Y/S})$.

When $S = \text{Spec } k$, $Y = \mathbb{A}_k^1$ and $Z = \{x\}$, then Conjecture 3.16 follows from Saito's isolated Milnor formula (3.15.1) and the cohomological Milnor formula (3.3.4).

Remark 3.17. Let $f : X \rightarrow Y$ be a separated morphism between smooth schemes over k , X is connected. Let $Z \subseteq X$ be a closed subset and $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Assume that f is universally locally acyclic relatively to $\mathcal{F}$ outside Z . Let $n = \dim X$. We expect that there is a n -cycle $CC_{X/Y}^Z(\mathcal{F})$ supported on $T^*X \times_X Z$ such that

$$(3.17.1) \quad \tilde{\text{cl}}(0_X^! CC_{X/Y}^Z(\mathcal{F})) = \tilde{C}_{X/Y/k}^Z(\mathcal{F}) \quad \text{in} \quad H_Z^0(X, \mathcal{K}_{X/Y/k}).$$

If Y is a smooth curve and Z is a finite set of closed points of X , then

$$(3.17.2) \quad CC_{X/Y}^Z(\mathcal{F}) = - \sum_{x \in Z} \dim_{\text{tot}} R\Phi_{\bar{x}}(\mathcal{F}, f) \cdot [T_x^* X].$$

If $f = \text{id}$ and Z is the smallest closed subset of X such that $\mathcal{F}|_{X \setminus Z}$ is smooth, then

$$(3.17.3) \quad CC_{X/Y}^Z(\mathcal{F}) = CC(\mathcal{F}) - \text{rank } \mathcal{F} \cdot CC(\Lambda).$$

Proposition 3.18. *Consider a Cartesian diagram in Sm_k*

$$(3.18.1) \quad \begin{array}{ccc} X' & \xrightarrow{i} & X \\ f' \downarrow & \square & \downarrow f \\ S' & \xrightarrow{\delta} & S. \end{array}$$

We assume that f and f' are smooth morphisms, and S and S' are connected of dimension s and s' respectively. Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$. Assume that $X \rightarrow S$ is $SS(\mathcal{F})$ -transversal and i is properly $SS(\mathcal{F})$ -transversal. Then we have

$$(3.18.2) \quad i^! cc_{X/S}(\mathcal{F}) = cc_{X'/S'}(i^* \mathcal{F}) \quad \text{in} \quad \text{CH}_{s'}(X'),$$

where $i^! : \text{CH}_s(X) \rightarrow \text{CH}_{s'}(X')$ is the refined Gysin pullback.

Proof. We consider the following diagram

$$(3.18.3) \quad \begin{array}{ccccc} & T^*X' & \xleftarrow{di} & T^*X \times_X X' & \xrightarrow{\text{pr}} & T^*X \\ & \downarrow & & \downarrow & & \downarrow \\ T^*S' \times_{S'} X' & \xleftarrow{d\delta} & T^*S \times_S X' & \xrightarrow{\quad} & T^*S \times_S X & \xrightarrow{\quad} & T^*X \\ & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ & T^*(X'/S') & \xrightarrow{\cong} & T^*(X/S) \times_X X' & \xrightarrow{\quad} & T^*(X/S) & \\ & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ X' & \xrightarrow{0_{X'/S'}} & X' & \xrightarrow{0_{X/S} \times 1} & X & \xrightarrow{0_{X/S}} & X \end{array}$$

Note that the square containing the morphisms di and $d\delta$ is Cartesian. In the following calculations, although di and $d\delta$ are not proper, we can still apply di_* and $d\delta_*$ since di is finite on the support of $\text{pr}^{-1}(SS(\mathcal{F}))$ and $d\delta$ is finite on the zero section X' of $T^*S \times_S X'$. We have

$$\begin{aligned}
 (3.18.4) \quad cc_{X'/S'}(i^*\mathcal{F}) &= (-1)^{s'} \cdot 0_{X'/S'}^! CC(i^*\mathcal{F}) \\
 &\stackrel{(a)}{=} (-1)^{s'} \cdot 0_{X'/S'}^! (di_* \text{pr}^! CC(\mathcal{F}) \cdot (-1)^{-\dim(X')+\dim(X)}) \\
 &= d\delta_* 0_{X'/S'}^! \text{pr}^! CC(\mathcal{F}) \cdot (-1)^{s'-\dim(X')+\dim(X)} \\
 &\stackrel{(b)}{=} 0_{X'/S'}^! \text{pr}^! CC(\mathcal{F}) \cdot (-1)^s = 0_{X/S}^! \text{pr}^! CC(\mathcal{F}) \cdot (-1)^s \\
 &= i^!(0_{X/S}^! CC(\mathcal{F}) \cdot (-1)^s) = i^! cc_{X/S}(\mathcal{F}).
 \end{aligned}$$

where (a) follows from [13, Theorem 7.6] and (b) follows from the fact that $0_{X'/S'}^! \text{pr}^! CC(\mathcal{F})$ is supported on the zero section of $T^*S \times_S X'$. $\square$

3.19. Consider a commutative diagram in Sm_k :

$$(3.19.1) \quad \begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow h & \swarrow g \\ & S & \end{array}$$

Let $\mathcal{F} \in D_{\text{ctf}}(X, \Lambda)$ such that h is $SS(\mathcal{F})$ -transversal. Assume f is proper on $\text{supp}(\mathcal{F})$. By [13, Lemma 3.8], g is $f_* SS(\mathcal{F})$ -transversal. By [4, Lemma 2.2(ii)], $SS(Rf_* \mathcal{F}) \subseteq f_* SS(\mathcal{F})$. Thus $g : Y \rightarrow S$ is also $SS(Rf_* \mathcal{F})$ -transversal and the class $cc_{Y/S}(Rf_* \mathcal{F})$ is well-defined.

Proposition 3.20. *Consider the assumptions in section 3.19. Assume moreover that Y is projective, $f : X \rightarrow Y$ is quasi-projective and $\dim f_* SS(\mathcal{F}) \leq \dim Y$. Then we have*

$$(3.20.1) \quad f_* cc_{X/S}(\mathcal{F}) = cc_{Y/S}(Rf_* \mathcal{F}) \quad \text{in} \quad \text{CH}_s(Y).$$

Proof. Consider the following commutative diagram

$$(3.20.2) \quad \begin{array}{ccccc} X & \xrightarrow{f} & Y & & \\ \downarrow 0_X & & \downarrow 0_Y & & \\ T^*S \times_S X & \xrightarrow{\text{id} \times f} & T^*S \times_S Y & & \\ \swarrow dh & \downarrow r & \downarrow dg & & \\ T^*X & \xleftarrow{df} T^*Y \times_Y X & \xrightarrow{\text{id} \times f} & T^*Y & \end{array}$$

Then we have

$$\begin{aligned}
 (3.20.3) \quad f_* cc_{X/S}(\mathcal{F}) &\stackrel{(3.5.2)}{=} (-1)^s \cdot f_* dh^! (CC(\mathcal{F})) = (-1)^s \cdot (\text{id} \times f)_* r^! df^! (CC(\mathcal{F})) \\
 &= (-1)^s \cdot dg^! (\text{id} \times f)_* df^! (CC(\mathcal{F})) \\
 &\stackrel{(a)}{=} (-1)^s \cdot dg^! CC(Rf_* \mathcal{F}) \stackrel{(3.5.2)}{=} cc_{Y/S}(Rf_* \mathcal{F}).
 \end{aligned}$$

where (a) follows from [14, Theorem 2.2.5]. $\square$

Lemma 3.21. *Consider the assumptions in section 3.8. On an open neighbourhood of $\text{supp}(\mathcal{F})$, the morphism $h : X \rightarrow S$ is smooth. Let*

$$(3.21.1) \quad \pi_{X/S} : T^*X \longrightarrow T^*(X/S), \quad q_S : \mathbb{P}(f^* T^*(Y/S)) \longrightarrow X$$

be the natural projections on this neighbourhood. The restriction of $\pi_{X/S}$ to $SS(\mathcal{F})$ is finite. The relative differential $\mathcal{O}(-1) \rightarrow q_S^* T^*(X/S)$ defines a section

$$(3.21.2) \quad \overline{d_{/S}f} : \mathbb{P}(f^* T^*(Y/S)) \longrightarrow q_S^* T^*(X/S) \otimes \mathcal{O}(1),$$

and we have

$$(3.21.3) \quad cc_{X/Y/S}^Z(\mathcal{F}) = (-1)^s (q_S)_* \left[c_{r-1} \left(\frac{q_S^* f^* T^*(Y/S)}{\mathcal{O}(-1)} \right) \cap \overline{d_{/S}f}^! ((\pi_{X/S})_* CC(\mathcal{F}))_{\mathcal{O}(1)} \right].$$

Here $(\pi_{X/S})_*$ is taken on the characteristic-cycle support, and $(q_S)_*$ is taken on the reduced intersection support over Z .

Proof. The morphism h is smooth on an open neighbourhood of $\text{supp}(\mathcal{F})$ by [13, Lemmas 3.6(6) and 4.2(3)], and we have

$$(3.21.4) \quad 0 \longrightarrow T^* S \times_S X \xrightarrow{dh} T^* X \xrightarrow{\pi_{X/S}} T^*(X/S) \longrightarrow 0.$$

on the open neighbourhood. Moreover,

$$(3.21.5) \quad SS(\mathcal{F}) \cap \ker(\pi_{X/S}) \subset 0(X),$$

so $\pi_{X/S}|_{SS(\mathcal{F})}$ is finite by [4, proof of Lemma 1.2(ii)].

The following square is Cartesian, since its vertical kernels are identified by dh :

$$(3.21.6) \quad \begin{array}{ccc} T^* Y \times_Y X & \xrightarrow{df} & T^* X \\ \downarrow & & \downarrow \pi_{X/S} \\ f^* T^*(Y/S) & \xrightarrow{d_{/S}f} & T^*(X/S). \end{array}$$

The tautological line bundle therefore gives

$$(3.21.7) \quad \mathcal{O}(-1) \longrightarrow q_S^* f^* T^*(Y/S) \longrightarrow q_S^* T^*(X/S),$$

hence we have a section $\overline{d_{/S}f}$ as in the statement. Since $\pi_{X/S}$ is finite on $SS(\mathcal{F})$, Gysin pullback commutes with proper pushforward [9, Theorem 6.2], hence we have

$$(3.21.8) \quad \rho_* df^! CC(\mathcal{F}) = (d_{/S}f)^! (\pi_{X/S})_* CC(\mathcal{F}).$$

By transversality,

$$(3.21.9) \quad (df)^{-1} SS(\mathcal{F}) \cap (T^* S \times_S X) \subset 0(X),$$

hence $T^* Y \times_Y X \longrightarrow f^* T^*(Y/S)$ induces a finite map

$$(3.21.10) \quad \bar{\rho} : \mathbb{P}((df)^{-1} SS(\mathcal{F})) \longrightarrow \mathbb{P}((d_{/S}f)^{-1} (\pi_{X/S} SS(\mathcal{F}))),$$

and $\bar{\rho}^* \mathcal{O}(-1) = \mathcal{O}(-1)$. By the compatibility with proper pushforward of Gysin pullback we have

$$(3.21.11) \quad \bar{\rho}_* (\bar{df}^! CC(\mathcal{F}))_{\mathcal{O}(1)} = \overline{d_{/S}f}^! ((\pi_{X/S})_* CC(\mathcal{F}))_{\mathcal{O}(1)},$$

hence

$$(3.21.12) \quad \bar{\rho}^* \left(\frac{q_S^* f^* T^*(Y/S)}{\mathcal{O}(-1)} \right) = \frac{q^* f^* T^*(Y/S)}{\mathcal{O}(-1)},$$

and the projection formula and (3.9.1) therefore implies

$$(3.21.13) \quad cc_{X/Y/S}^Z(\mathcal{F}) = (-1)^s (q_S)_* \left[c_{r-1} \left(\frac{q_S^* f^* T^*(Y/S)}{\mathcal{O}(-1)} \right) \cap \overline{d_{/S}f}^! ((\pi_{X/S})_* CC(\mathcal{F}))_{\mathcal{O}(1)} \right].$$

which is (3.21.3). $\square$

Proposition 3.22. *Consider a commutative diagram in Sm_k*

$$(3.22.1) \quad \begin{array}{ccc} X' & \xrightarrow{i_X} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{i_Y} & Y \\ g' \downarrow & & \downarrow g \\ S' & \xrightarrow{\delta} & S, \end{array}$$

whose squares are Cartesian. Put $Z' = Z \times_X X'$, and write $s' = \dim S'$. Assume that X', Y', S' are smooth and pure-dimensional, that S' is connected, that $\dim X' - \dim X = s' - s$, and that i_X is properly $SS(\mathcal{F})$ -transversal. Then

$$(3.22.2) \quad i_X^! cc_{X/Y/S}^Z(\mathcal{F}) = cc_{X'/Y'/S'}^{Z'}(i_X^* \mathcal{F}) \quad \text{in } \text{CH}_{s'}(Z').$$

Proof. Using the Cartesian diagram (3.22.1), proper $SS(\mathcal{F})$ -transversality gives

$$(3.22.3) \quad SS(i_X^* \mathcal{F}) \subset di_X(\text{pr}^{-1} SS(\mathcal{F})), \quad \text{pr} : T^* X \times_X X' \longrightarrow T^* X.$$

Hence

$$(3.22.4) \quad SS(i_X^* \mathcal{F}) \cap (T^* S' \times_{S'} X') \subset 0(X'), \quad (df')^{-1} SS(i_X^* \mathcal{F}) \subset 0(X') \cup (T^* Y' \times_{Y'} Z').$$

Thus $g' \circ f'$ is $SS(i_X^* \mathcal{F})$ -transversal and $f'|_{X' \setminus Z'}$ is $SS((i_X^* \mathcal{F})|_{X' \setminus Z'})$ -transversal. Pullback formula of characteristic cycles [13, Theorem 7.6] gives

$$(3.22.5) \quad CC(i_X^* \mathcal{F}) = (-1)^{\dim X' - \dim X} (di_X)_* \text{pr}^! CC(\mathcal{F}),$$

then by base change we have

$$(3.22.6) \quad (\pi_{X'/S'})_* CC(i_X^* \mathcal{F}) = (-1)^{\dim X' - \dim X} i_{X/S}^! (\pi_{X/S})_* CC(\mathcal{F}),$$

where

$$(3.22.7) \quad i_{X/S} : T^*(X'/S') \simeq T^*(X/S) \times_X X' \longrightarrow T^*(X/S)$$

is the natural projection. Let $i_{\mathbb{P}} : \mathbb{P}(f'^* T^*(Y'/S')) \longrightarrow \mathbb{P}(f^* T^*(Y/S))$ be the induced morphism, then

$$(3.22.8) \quad \begin{array}{ccc} \mathbb{P}(f'^* T^*(Y'/S')) & \xrightarrow{i_{\mathbb{P}}} & \mathbb{P}(f^* T^*(Y/S)) \\ q_{S'} \downarrow & & \downarrow q_S \\ X' & \xrightarrow{i_X} & X \end{array}$$

is Cartesian, and $i_{\mathbb{P}}^* \mathcal{O}(-1) = \mathcal{O}(-1)$, $i_{\mathbb{P}}^* q_S^* f^* T^*(Y/S) = q_{S'}^* f'^* T^*(Y'/S')$. Thus

$$(3.22.9) \quad i_{\mathbb{P}}^* \left(\frac{q_S^* f^* T^*(Y/S)}{\mathcal{O}(-1)} \right) = \frac{q_{S'}^* f'^* T^*(Y'/S')}{\mathcal{O}(-1)}.$$

The relative differentials commute with base change, giving the following Cartesian diagram:

$$(3.22.10) \quad \begin{array}{ccc} \mathbb{P}(f'^* T^*(Y'/S')) & \xrightarrow{i_{\mathbb{P}}} & \mathbb{P}(f^* T^*(Y/S)) \\ \overline{d_{/S'} f'} \downarrow & & \downarrow \overline{d_{/S} f} \\ q_{S'}^* T^*(X'/S') \otimes \mathcal{O}(1) & \longrightarrow & q_S^* T^*(X/S) \otimes \mathcal{O}(1). \end{array}$$

Here the lower horizontal arrow is the natural projection under the identification $q_{S'}^* T^*(X'/S') \otimes \mathcal{O}(1) \simeq i_{\mathbb{P}}^*(q_S^* T^*(X/S) \otimes \mathcal{O}(1))$. Together with (3.22.6), we have

$$(3.22.11) \quad \overline{d_{/S'}} f'^1 ((\pi_{X'/S'})_* CC(i_X^* \mathcal{F}))_{\mathcal{O}(1)} = (-1)^{\dim X' - \dim X} i_{\mathbb{P}}^! \left[\overline{d_{/S}} f^1 ((\pi_{X/S})_* CC(\mathcal{F}))_{\mathcal{O}(1)} \right].$$

Applying (3.21.3) and $i_X^! (q_S)_* = (q_{S'})_* i_{\mathbb{P}}^!$ we have

$$(3.22.12) \quad cc_{X'/Y'/S'}^{Z'}(i_X^* \mathcal{F}) = (-1)^{s' + \dim X' - \dim X - s} i_X^! cc_{X/Y/S}^Z(\mathcal{F}).$$

Since $\dim X' - \dim X = s' - s$, we have (3.22.2). $\square$

In particular, if i_X is flat of pure relative dimension $s' - s$, then

$$(3.22.13) \quad \text{pr}^{-1} SS(\mathcal{F}) \cap \ker(di_X) \subset 0(X'),$$

so i_X is $SS(\mathcal{F})$ -transversal; moreover, for every irreducible component $C \subset SS(\mathcal{F})$,

$$(3.22.14) \quad \dim(C \times_X X') = \dim C + s' - s = \dim X + s' - s = \dim X',$$

hence i_X is properly $SS(\mathcal{F})$ -transversal and the pullback formula applies.

Proposition 3.23. *Let $p : X \rightarrow X'$ be a proper morphism in Sm_k , with X' of pure dimension, and let $f' : X' \rightarrow Y$ satisfy $f = f' \circ p$. Let $Z' \subseteq X'$ be closed, with $Z \subseteq p^{-1}(Z')$. Assume that $\dim p_* SS(\mathcal{F}) \leq \dim X'$, then $g \circ f'$ is $SS(Rp_* \mathcal{F})$ -transversal, and f' is $SS(Rp_* \mathcal{F})$ -transversal outside Z' . Moreover,*

$$(3.23.1) \quad p_* cc_{X/Y/S}^Z(\mathcal{F}) = cc_{X'/Y/S}^{Z'}(Rp_* \mathcal{F}) \quad \text{in } \text{CH}_s(Z').$$

Proof. Proper pushforward of singular support gives $SS(Rp_* \mathcal{F}) \subseteq p_* SS(\mathcal{F})$ [4, Lemma 2.2(ii)]. A covector in $p_* SS(\mathcal{F})$ lifts to a covector whose pullback by dp belongs to $SS(\mathcal{F})$. The hypotheses on h and f therefore give

$$(3.23.2) \quad \begin{aligned} (df')^{-1} p_* SS(\mathcal{F}) &\subseteq 0(X') \cup (T^* Y \times_Y Z'), \\ (df')^{-1} p_* SS(\mathcal{F}) \cap (T^* S \times_S X') &\subseteq 0(X'). \end{aligned}$$

Here $0(X')$ is the zero section of $T^* Y \times_Y X'$. This proves both transversality assertions, and also gives a common projective support on which the quotient bundle is defined.

Abe's proper-pushforward theorem [2, Introduction, Theorem; §6.6] gives

$$(3.23.3) \quad CC(Rp_* \mathcal{F}) = p_! CC(\mathcal{F}) = \text{pr}_{2*} (dp)^! CC(\mathcal{F}) \quad \in \text{CH}_{\dim X'}(p_* SS(\mathcal{F})) \left[\frac{1}{\text{char } k} \right]$$

where $\text{char } k > 0$. Let q' be the projective projection over X' , and let

$$(3.23.4) \quad p_{\mathbb{P}} : \mathbb{P}(T^* Y \times_Y X) \longrightarrow \mathbb{P}(T^* Y \times_Y X')$$

be the base change of p . We have natural maps

$$(3.23.5) \quad \begin{aligned} dp \otimes 1 : q^*(T^* X' \times_{X'} X) \otimes \mathcal{O}(1) &\longrightarrow q^* T^* X \otimes \mathcal{O}(1), \\ \rho : q^*(T^* X' \times_{X'} X) \otimes \mathcal{O}(1) &\longrightarrow q^* T^* X' \otimes \mathcal{O}(1). \end{aligned}$$

The second map is proper. Let

$$(3.23.6) \quad D \subset q'^* T^* X' \otimes \mathcal{O}(1)$$

be the $\mathcal{O}(1)$ -twist of the pullback of $p_* SS(\mathcal{F})$, then $\dim D \leq \dim X' + \dim Y - 1$. On an open subset trivializing $\mathcal{O}(-1)$ we have

$$(3.23.7) \quad \rho_*(dp \otimes 1)^! CC(\mathcal{F})_{\mathcal{O}(1)} = CC(Rp_* \mathcal{F})_{\mathcal{O}(1)} \quad \text{in } \text{CH}_{\dim X' + \dim Y - 1}(D) \left[\frac{1}{\text{char } k} \right].$$

Since $\mathrm{CH}_{\dim X' + \dim Y - 1}(D) = Z_{\dim X' + \dim Y - 1}(D)$ and inverting $\mathrm{char} k$ is injective on the free abelian group $Z_{\dim X' + \dim Y - 1}(D)$, the local equalities (3.23.7) are equalities of cycles and hence glue globally into an equality in $Z_{\dim X' + \dim Y - 1}(D)$. Let $p_{\mathbb{P}}$ be the base change of p . The pullback of $\overline{df}'$ is a section

$$(3.23.8) \quad p_{\mathbb{P}}^* \overline{df} : \mathbb{P}(T^*Y \times_Y X) \longrightarrow q^*(T^*X' \times_{X'} X) \otimes \mathcal{O}(1),$$

and

$$(3.23.9) \quad (dp \otimes 1) \circ p_{\mathbb{P}}^* \overline{df}' = \overline{df}, \quad \rho \circ p_{\mathbb{P}}^* \overline{df}' = \overline{df}' \circ p_{\mathbb{P}}.$$

Therefore, by compatibility of refined Gysin pullback with proper pushforward [9, Theorem 6.2],

$$(3.23.10) \quad \begin{aligned} (p_{\mathbb{P}})_* \overline{df}^! CC(\mathcal{F})_{\mathcal{O}(1)} &= (p_{\mathbb{P}})_* (p_{\mathbb{P}}^* \overline{df}')^! (dp \otimes 1)^! CC(\mathcal{F})_{\mathcal{O}(1)} \\ &= \overline{df}'^! \rho_* (dp \otimes 1)^! CC(\mathcal{F})_{\mathcal{O}(1)} \\ &= \overline{df}'^! CC(Rp_* \mathcal{F})_{\mathcal{O}(1)}. \end{aligned}$$

Combining with

$$(3.23.11) \quad p_{\mathbb{P}}^* \left(\frac{q'^* f'^* T^*(Y/S)}{\mathcal{O}(-1)} \right) = \frac{q^* f^* T^*(Y/S)}{\mathcal{O}(-1)},$$

we have

$$(3.23.12) \quad \begin{aligned} p_* cc_{X/Y/S}^Z(\mathcal{F}) &= (-1)^s q'_* \left[c_{r-1} \left(\frac{q'^* f'^* T^*(Y/S)}{\mathcal{O}(-1)} \right) \cap \overline{df}'^! CC(Rp_* \mathcal{F})_{\mathcal{O}(1)} \right] \\ &= cc_{X'/Y/S}^{Z'}(Rp_* \mathcal{F}), \end{aligned}$$

which proves the assertion. $\square$

Corollary 3.24 (Conductor formula). *Let $S = \mathrm{Spec}(k)$, let Y be a smooth connected curve, and let $f : X \rightarrow Y$ be proper. Let $T \subset Y$ be a finite set of closed points, and put $Z = f^{-1}(T)$. Assume that f is $SS(\mathcal{F})$ -transversal outside Z . Then*

$$(3.24.1) \quad f_* cc_{X/Y/k}^Z(\mathcal{F}) = - \sum_{y \in T} a_y (Rf_* \mathcal{F})[y] \quad \text{in } \mathrm{CH}_0(T),$$

where a_y is the Artin conductor, computed after derived reduction of the coefficient ring to its residue field.

Proof. Combining Theorem 3.15 and Proposition 3.23 we get the desired result. $\square$

4. NON-ISOLATED MILNOR FORMULA IN QUASI-AFFINE CASE

4.1. Let k be an algebraically closed field of characteristic $p > 0$, Λ be a finite local ring such that the characteristic of the residue field is invertible in k . Let X be a smooth separated scheme of finite type over k purely of dimension d , let $f : X \rightarrow \mathbb{A}_k^1$ be a morphism and let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. Let t be the standard coordinate on $\mathbb{A}_k^1$. The form dt defines

$$(4.1.1) \quad df : X \longrightarrow X \times_{\mathbb{A}_k^1} T^* \mathbb{A}_k^1 \longrightarrow T^* X$$

which gives a section of T^*X . Put $Z = Z_{X,f} := (df)^{-1} SS(\mathcal{F})$, [13, Lemma 4.2(5)] implies f is universally locally acyclic relatively to $\mathcal{F}$ in $X \setminus Z$.

Theorem 4.2. *Consider the previous assumptions. Assume that X is quasi-affine. Then we have*

$$(4.2.1) \quad \tilde{C}_{X/\mathbb{A}_k^1/k}^Z(\mathcal{F}) = \tilde{\mathrm{cl}} \left(df^! CC(\mathcal{F}) \right).$$

By Proposition 3.14, $cc_{X/\mathbb{A}_k^1/k}^Z(\mathcal{F}) = df^!CC(\mathcal{F})$ in $\mathrm{CH}_0(Z)$. Hence Theorem 4.2 verifies Conjecture 3.16 in the quasi-affine case, with $Y = \mathbb{A}_k^1$, $S = \mathrm{Spec} k$.

We devote the rest of this section to the proof of the theorem.

4.3. We first recall Fulton's construction of specialization over a discrete valuation ring [9, Chapter 20]. Let R be a discrete valuation ring, let $S = \mathrm{Spec} R$ with $j : \eta \hookrightarrow S$, $i : s \hookrightarrow S$ be its generic point and closed point, respectively. Let u be a uniformizer of R .

Define the dimension function by setting $\delta_S(s) = 0$, $\delta_S(\eta) = 1$ and

$$(4.3.1) \quad \delta_T(x) = \delta_S(p(x)) + \mathrm{trdeg}_{\kappa(p(x))} \kappa(x)$$

for a morphism $p : T \rightarrow S$ locally of finite type with $x \in T$. For an integral closed subscheme $V \subset T$, its S -dimension is defined to be $\delta_T(\xi_V)$ where ξ_V is the generic point of V . Let $Z_m(T/S)$ be the free abelian group on integral closed subschemes of S -dimension m , we may define rational equivalence in the usual way: it is generated by $\mathrm{div}_V(h)$ for rational functions h on integral closed V of S -dimension $m+1$. Put $\mathrm{CH}_m(T/S) = Z_m(T/S)/\sim_{\mathrm{rat}}$. Let $j_T : T_\eta \rightarrow T$, $i_T : T_s \rightarrow T$ be the base changes of j and i , respectively. Then, with a slight abuse of notation, we have

$$(4.3.2) \quad j_T^* : \mathrm{CH}_{m+1}(T/S) \longrightarrow \mathrm{CH}_m(T_\eta), \quad i_T^! : \mathrm{CH}_{m+1}(T/S) \longrightarrow \mathrm{CH}_m(T_s).$$

The second map is the refined Gysin map obtained by the regular immersion i . Moreover, we have the localization exact sequence

$$(4.3.3) \quad \mathrm{CH}_{r+1}(T_s) \xrightarrow{i_{T*}} \mathrm{CH}_{r+1}(T/S) \xrightarrow{j_T^*} \mathrm{CH}_r(T_\eta) \longrightarrow 0.$$

For any $\alpha \in \mathrm{CH}_r(T_\eta)$, choose a lift $\tilde{\alpha} \in \mathrm{CH}_{r+1}(T/S)$ by (4.3.3) and define

$$(4.3.4) \quad \mathrm{sp}^{\mathrm{cyc}}(\alpha) = i_T^!(\tilde{\alpha}) \in \mathrm{CH}_r(T_s).$$

One may check $\mathrm{sp}^{\mathrm{cyc}}(\alpha)$ is independent of the lift. Indeed, two lifts differ by some $i_{T*}\xi$, and the refined self-intersection formula gives

$$(4.3.5) \quad i_T^! i_{T*} \xi = c_1(p_s^* N_{s/S}) \cap \xi = 0,$$

because $N_{s/S} \simeq \mathcal{O}_s$. In particular,

$$(4.3.6) \quad \mathrm{sp}^{\mathrm{cyc}}(j_T^* \gamma) = i_T^! \gamma$$

for $\gamma \in \mathrm{CH}_{r+1}(T/S)$. For integral subscheme $V \subset T$ such that $V_\eta \neq \emptyset$ (in this case we call V *horizontal*), $\mathrm{sp}^{\mathrm{cyc}}([V_\eta])$ is the divisor $\mathrm{div}_V(u)$, with the usual length multiplicities along the special fiber.

4.4. Let B be a smooth scheme separated of finite type over S . Let $E \rightarrow B$ be a vector bundle of rank r , let $\sigma : B \rightarrow E$ be a section, let C be a closed subscheme of E , and put $Z = B \times_E C$. Since σ is a regular immersion of codimension r , there is a supported refined Gysin map

$$(4.4.1) \quad \sigma^! : \mathrm{CH}_{m+1}(C/S) \longrightarrow \mathrm{CH}_{m-r+1}(Z/S).$$

Lemma 4.5. *Consider the assumptions in §4.4. Let $\gamma \in \mathrm{CH}_{m+1}(C/S)$. Define*

$$(4.5.1) \quad \gamma_\eta = j_C^* \gamma \in \mathrm{CH}_m(C_\eta), \quad \gamma_s = i_C^! \gamma \in \mathrm{CH}_m(C_s).$$

Then we have

$$(4.5.2) \quad j_Z^*(\sigma^! \gamma) = \sigma_\eta^! \gamma_\eta \quad \text{in } \mathrm{CH}_{m-r}(Z_\eta),$$

$$(4.5.3) \quad \mathrm{sp}^{\mathrm{cyc}}(\sigma_\eta^! \gamma_\eta) = \sigma_s^! \gamma_s \quad \text{in } \mathrm{CH}_{m-r}(Z_s).$$

Proof. Refined Gysin pullback exchanges with flat base change $\eta \rightarrow S$, hence we have

$$(4.5.4) \quad j_Z^* \circ \sigma^! = \sigma_\eta^! \circ j_C^*,$$

which proves (4.5.2).

For the special fiber, we have

$$(4.5.5) \quad \mathrm{sp}^{\mathrm{cyc}}(\sigma_\eta^! \gamma_\eta) \stackrel{(4.5.2)}{=} \mathrm{sp}^{\mathrm{cyc}}(j_Z^* \sigma^! \gamma) \stackrel{(4.3.6)}{=} i_Z^! \sigma^! \gamma \stackrel{[9, \S 6.4]}{=} \sigma_s^! i_C^! \gamma.$$

□

Corollary 4.6. *Consider the assumptions in Lemma 4.5. Suppose*

$$(4.6.1) \quad \alpha = \sum_a n_a [C_a]$$

is a cycle supported on C , with components $C_a \subset E$ flat over S of pure relative dimension m , and put $q = m - r \geq 0$. For $t \in \{\eta, s\}$ define

$$(4.6.2) \quad \alpha_t := \sum_{a: (C_a)_t \neq \emptyset} n_a [(C_a)_t],$$

with the usual scheme-theoretic multiplicities at its generic points. Then

$$(4.6.3) \quad \mathrm{sp}^{\mathrm{cyc}}(\sigma_\eta^! \alpha_\eta) = \sigma_s^! \alpha_s \quad \text{in } \mathrm{CH}_q(Z_s).$$

Proof. Since each C_a is flat over S with pure m -dimension fibers, we have a natural lift $\tilde{\alpha} = \sum_a n_a [C_a]$ in $\mathrm{CH}_{m+1}(C/S)$. Flat pullback gives $j_C^* \tilde{\alpha} = \alpha_\eta$, while Gysin pullback along $s \rightarrow S$ gives $i_C^! \tilde{\alpha} = \alpha_s$. The consequence follows from Lemma 4.5. □

4.7. In sections 4.7–4.12, let $S = \mathrm{Spec} R$ be a strictly henselian trait with generic point η and closed point s , and let $u \in R$ be a uniformizer. Let $f : X \rightarrow S$ be a smooth separated morphism of finite type and pure relative dimension d . Let $k : Z \rightarrow X$ be a closed subscheme. Let $j_X : X_\eta \rightarrow X$, $i_X : X_s \rightarrow X$ be the generic and special fiber, respectively, and $b_X : X_{\bar{\eta}} \rightarrow X_\eta$ be the base change to the algebraic closure $\bar{\eta}$. We use the convention of nearby cycle functor in [18, §3.4]:

$$(4.7.1) \quad R\Psi_f = i_X^* Rj_{X*} b_{X*} b_X^* : D^+(X_\eta, \Lambda) \rightarrow D^+(X_s, \Lambda).$$

For $0 \leq r \leq d$, $c = d - r$ and $t \in \{\eta, s\}$, we have the refined cycle class map

$$(4.7.2) \quad \mathrm{cl}_t : \mathrm{CH}_r(Z_t) \longrightarrow H_{Z_t}^{2c}(X_t, \Lambda(c)) \simeq H_{Z_t}^{-2r}(X_t, \mathcal{K}_{X_t/t}(-r))$$

by Riou [12, Exposé XVI, Definition 2.3.1, Proposition 2.3.7, Definition 2.4.1].

Using the method of [18, §3.5], we have a natural specialization map on étale cohomology groups

$$(4.7.3) \quad \mathrm{sp}^{\mathrm{ét}} : H_{Z_\eta}^{-2r}(X_\eta, \mathcal{K}_{X_\eta/\eta}(-r)) \longrightarrow H_{Z_s}^{-2r}(X_s, \mathcal{K}_{X_s/s}(-r)).$$

Theorem 4.8. *For every $0 \leq r \leq d$ and $c = d - r$, the square*

$$(4.8.1) \quad \begin{array}{ccc} \mathrm{CH}_r(Z_\eta) & \xrightarrow{\mathrm{cl}_\eta} & H_{Z_\eta}^{2c}(X_\eta, \Lambda(c)) \\ \mathrm{sp}^{\mathrm{cyc}} \downarrow & & \downarrow \mathrm{sp}^{\mathrm{ét}} \\ \mathrm{CH}_r(Z_s) & \xrightarrow{\mathrm{cl}_s} & H_{Z_s}^{2c}(X_s, \Lambda(c)) \end{array}$$

commutes.

We postpone the proof of Theorem 4.8 to §4.11.

Lemma 4.9. *Let $W \subset Z$ be closed and $\gamma \in H_W^{2a}(X, \mathcal{K}_{X/S}(a))$, we have*

$$(4.9.1) \quad \mathrm{sp}^{\mathrm{ét}}(j_X^* \gamma) = i_X^* \gamma \quad \text{in} \quad H_{W_s}^{2a}(X_s, \mathcal{K}_{X_s/s}(a)).$$

Proof. Let $\ell : W \rightarrow X$ be the closed immersion. By definition, γ is equivalent to a morphism $\gamma : \ell_* \Lambda_W \rightarrow \mathcal{K}_{X/S}(a)[2a]$ and we have

$$(4.9.2) \quad \ell_{s*} \Lambda_{W_s} \simeq i_X^* \ell_* \Lambda_W \longrightarrow R\Psi_f j_X^* \ell_* \Lambda_W \simeq R\Psi_f(\ell_{\eta*} \Lambda_{W_\eta}),$$

$$(4.9.3) \quad i_X^* \mathcal{K}_{X/S}(a)[2a] \longrightarrow R\Psi_f(j_X^* \mathcal{K}_{X/S})(a)[2a] \simeq R\Psi_f(\mathcal{K}_{X_\eta/\eta})(a)[2a].$$

by proper base change. Moreover, we have the following commutative square

$$(4.9.4) \quad \begin{array}{ccc} \ell_{s*} \Lambda_{W_s} & \xrightarrow{(4.9.2)} & R\Psi_f(\ell_{\eta*} \Lambda_{W_\eta}) \\ i_X^* \gamma \downarrow & & \downarrow R\Psi_f(j_X^* \gamma) \\ i_X^* \mathcal{K}_{X/S}(a)[2a] & \xrightarrow{(4.9.3)} & R\Psi_f(\mathcal{K}_{X_\eta/\eta})(a)[2a] \end{array}$$

by the naturality of the nearby cycle functor. By [18, (3.67)], we have a natural map

$$(4.9.5) \quad \theta_f : R\Psi_f(\mathcal{K}_{X_\eta/\eta}) \longrightarrow \mathcal{K}_{X_s/s}.$$

Combining the definition of specialization in [18, (3.68), (3.69)] with (4.9.4), we have

$$(4.9.6) \quad \mathrm{sp}^{\mathrm{et}}(j_X^* \gamma) = \theta_f(a)[2a] \circ R\Psi_f(j_X^* \gamma) \circ (4.9.2) = \theta_f(a)[2a] \circ (4.9.3) \circ i_X^* \gamma.$$

Since f is smooth, local acyclicity and purity imply $\theta_f(a)[2a] \circ (4.9.3)$ is precisely the isomorphism

$$(4.9.7) \quad i_X^* \mathcal{K}_{X/S}(a)[2a] \xrightarrow{\sim} \mathcal{K}_{X_s/s}(a)[2a],$$

hence the right hand side of (4.9.6) is exactly $i_X^* \gamma$. □

4.10. We use the following result of SGA 4 $\frac{1}{2}$ [7]. Assume X' is smooth over S of pure relative dimension N , a closed subscheme $Z' \subset X'$ has fiber codimension at least c and $\mathcal{K}$ is a bounded coherent perfect $\mathcal{O}_{X'}$ -complex supported on Z' and of finite Tor-dimension over S . Then [7, Cycle, 2.3.3] constructs a general cycle class

$$(4.10.1) \quad \mathrm{cl}_S(Z', \mathcal{K}) \in H_{Z'}^{2c}(X', \Lambda(c)).$$

Its formation commutes with arbitrary base change, and when S is a field, it agrees with the usual supported cycle class and factors through the Chow group. See [7, Cycle, Lemme 2.3.7, 2.2.10].

4.11. We now prove Theorem 4.8.

Proof of Theorem 4.8. Both paths in (4.8.1) are homomorphisms on Chow groups, so it is enough to treat a prime cycle $[W_\eta]$ supported on Z .

Let $W \subset Z$ be the scheme-theoretic closure of W_η which is integral. Multiplication by u is injective in every affine coordinate ring of W , so W is torsion-free over $\mathrm{Spec} R$, hence flat over S by [15, Tag 0539]. If the special fiber W_s is nonempty, flatness implies $\dim W_s = r$. Set $c = d - r$, the codimension. We have

$$(4.11.1) \quad \mathrm{sp}^{\mathrm{cyc}}[W_\eta] = \sum_{\xi \in (W_s)^{(0)}} m_\xi [\overline{\{\xi\}}], \quad m_\xi = \mathrm{length}_{\mathcal{O}_{W,\xi}}(\mathcal{O}_{W,\xi}/u)$$

by the previous description of $\mathrm{sp}^{\mathrm{cyc}}([V])$ of a horizontal integral subscheme V . If W_s is empty, the right side of (4.11.1) is zero. In the following we assume $W_s \neq \emptyset$.

The scheme X is regular and Noetherian because S is a regular trait and f is smooth of finite type; see [15, Tags 0H7S and 0FDC]. Hence the coherent module $\mathcal{O}_W$ is perfect as an $\mathcal{O}_X$ -complex by [15, Tag 0FDC]. Moreover, flatness of W over S implies $\mathcal{O}_W$ is finite Tor-dimension over S . Thus by [7, Cycle, 2.3.3] we may form

$$(4.11.2) \quad \mathrm{cl}_S(W, \mathcal{O}_W) \in H_W^{2c}(X, \Lambda(c)) \simeq H_W^{-2r}(X, \mathcal{K}_{X/S}(-r)).$$

On the generic fiber, the length of $\mathcal{O}_{W_\eta}$ is one. On the special fiber, flatness gives $Li_X^* \mathcal{O}_W = \mathcal{O}_{W_s}$, whose generic length at ξ is the integer m_ξ in (4.11.1). Therefore by (4.11.1) and [7, Cycle, Lemme 2.3.7, Théorème 2.3.8(i)], we have

$$(4.11.3) \quad j_X^* \text{cl}_S(W, \mathcal{O}_W) = \text{cl}_\eta[W_\eta],$$

$$(4.11.4) \quad i_X^* \text{cl}_S(W, \mathcal{O}_W) = \sum_{\xi} m_\xi \text{cl}_s[\overline{\{\xi\}}] = \text{cl}_s(\text{sp}^{\text{cyc}}[W_\eta]).$$

Applying Lemma 4.9, we have

$$(4.11.5) \quad \text{sp}^{\text{ét}} \text{cl}_\eta[W_\eta] = \text{sp}^{\text{ét}}(j_X^* \text{cl}_S(W, \mathcal{O}_W)) = i_X^* \text{cl}_S(W, \mathcal{O}_W) = \text{cl}_s(\text{sp}^{\text{cyc}}[W_\eta]).$$

This proves the commutativity of (4.8.1) on prime cycles. Fulton [9, §20.3] proves that sp^{cyc} respects rational equivalence, and the supported cycle class map factors through Chow groups by [7, Cycle, §2.2]. The equality therefore descends to $\text{CH}_r(Z_\eta)$. $\square$

Corollary 4.12. *In the notation of Lemma 4.5, suppose in addition that S is a strictly henselian trait and $B \rightarrow S$ is smooth of pure relative dimension d . Put $0 \leq q = m - r \leq d$, then in $H_{Z_s}^{-2q}(B_s, \mathcal{K}_{B_s/s}(-q)) \simeq H_{Z_s}^{2(d-q)}(B_s, \Lambda(d-q))$ we have*

$$(4.12.1) \quad \text{sp}^{\text{ét}} \text{cl}_\eta(\sigma_\eta^! \gamma_\eta) = \text{cl}_s(\sigma_s^! \gamma_s).$$

More generally, let

$$(4.12.2) \quad \tilde{\alpha} \in Z_{m+1}(C/S)$$

be any cycle, denote its class in $\text{CH}_{m+1}(C/S)$ by γ , and set

$$(4.12.3) \quad \alpha_\eta := j_C^* \gamma \in \text{CH}_m(C_\eta), \quad \alpha_s := i_C^! \gamma \in \text{CH}_m(C_s).$$

Then (4.12.1) has the form

$$(4.12.4) \quad \text{sp}^{\text{ét}} \text{cl}_\eta(\sigma_\eta^! \alpha_\eta) = \text{cl}_s(\sigma_s^! \alpha_s).$$

Proof. Apply Theorem 4.8 to $\sigma_\eta^! \gamma_\eta$ we have

$$(4.12.5) \quad \text{sp}^{\text{ét}} \text{cl}_\eta(\sigma_\eta^! \gamma_\eta) = \text{cl}_s(\text{sp}^{\text{cyc}}(\sigma_\eta^! \gamma_\eta)).$$

Now use (4.5.3) we get the desired conclusion. $\square$

Proposition 4.13. *For a separated scheme X of finite type over k , the following conditions are equivalent:*

- (1) *there is a finite-dimensional subspace $V \subset \Gamma(X, \mathcal{O}_X)$ such that*

$$(4.13.1) \quad V \otimes_k \mathcal{O}_X \longrightarrow \Omega_{X/k}^1, \quad v \otimes g \longmapsto g \cdot dv,$$

is surjective.

- (2) *X is quasi-affine.*

Proof. First assume condition (2). By [15, Tag 04II], there is a closed immersion $i : X \rightarrow U$, where U is an open subscheme of $\mathbb{A}_k^n$. Therefore, the natural homomorphism $i^* \Omega_{U/k}^1 \rightarrow \Omega_{X/k}^1$ is a surjection. The consequence follows from

$$(4.13.2) \quad i^* \Omega_{U/k}^1 \simeq \bigoplus_{j=1}^n \mathcal{O}_X dx_j.$$

Now we show that (1) implies (2). Let V be the subspace of $\Gamma(X, \mathcal{O}_X)$ satisfying the condition, and fix a basis $g_1, \dots, g_N$. Put

$$(4.13.3) \quad g = (g_1, \dots, g_N) : X \rightarrow \mathbb{A}_k^N.$$

In the cotangent sequence

$$(4.13.4) \quad g^* \Omega_{\mathbb{A}_k^N/k}^1 \longrightarrow \Omega_{X/k}^1 \longrightarrow \Omega_{X/\mathbb{A}_k^N}^1 \longrightarrow 0,$$

the first arrow is $dx_i \mapsto dg_i$, where $x_1, \dots, x_N$ are the coordinates of $\mathbb{A}_k^N$. The surjectivity condition hence implies $\Omega_{X/\mathbb{A}_k^N}^1 = 0$ and g is unramified. An unramified morphism is locally quasi-finite; because g is also quasi-compact, it is quasi-finite. See [15, Tags 02V5 and 01TJ].

The diagonal map $X \rightarrow X \times_{\mathbb{A}_k^N} X$ is a base change of the diagonal $X \rightarrow X \times_k X$; hence g is separated. We may apply Zariski's main theorem to get the factorization

$$(4.13.5) \quad X \xrightarrow{j} \overline{X} \xrightarrow{\overline{g}} \mathbb{A}_k^N,$$

where j is an open immersion and $\overline{g}$ is finite. Hence $\overline{X}$ is affine, and its open subscheme X is quasi-affine. $\square$

4.14. In the following we use the notations of Theorem 4.2. Let V be a subspace of $\Gamma(X, \mathcal{O}_X)$ obtained by Proposition 4.13, with $N = \dim V$, $\mathbb{V} := \operatorname{Spec}(\operatorname{Sym}_k V^\vee) \simeq \mathbb{A}_k^N$. Define

$$(4.14.1) \quad a : X \times_k \mathbb{V} \longrightarrow T^*X, \quad (x, g) \longmapsto df_x + dg_x.$$

Lemma 4.15. *There is a dense open subset $\mathbb{V}^\circ \subset \mathbb{V}$ such that $(df + dg)^{-1}(SS(\mathcal{F}))$ is finite for every $g \in \mathbb{V}^\circ(k)$. One may choose $g \in \mathbb{V}(k)$ such that the generic point of the line $k \cdot g$ belongs to $\mathbb{V}^\circ$.*

Proof. The map a lies over the identity of X , and its vertical differential is calculated by the surjective evaluation $V \rightarrow T_x^*X$, $g \mapsto dg_x$. Hence a is smooth of relative dimension $N - d$. Since each component of $SS(\mathcal{F})$ has dimension d , every component of

$$(4.15.1) \quad Q = a^{-1}(SS(\mathcal{F})) \subset X \times \mathbb{V}$$

has dimension N . Let $q : Q \rightarrow \mathbb{V}$ be the projection, let $Q_1, \dots, Q_m$ be the irreducible components of Q . Put

$$(4.15.2) \quad D := \bigcup_{\overline{q(Q_i)} \subsetneq \mathbb{V}} \overline{q(Q_i)} \subsetneq \mathbb{V},$$

$U_D := \mathbb{V} \setminus D$ is the complement. Then for any $v \in U_D$, and Q_i with $\overline{q(Q_i)} \subsetneq \mathbb{V}$, we have $(Q_i)_v = \emptyset$.

For Q_j such that Q_j dominates $\mathbb{V}$, i.e. $\overline{q(Q_j)} = \mathbb{V}$, we have

$$(4.15.3) \quad \dim(Q_j)_\eta = \dim Q_j - \dim \mathbb{V} = 0$$

where η is the generic point of $\mathbb{V}$. By a generic fiber dimension result [15, Tag 05F7], $\dim(Q_j)_u = 0$ for all u in an open subscheme U_j of $\mathbb{V}$. Therefore,

$$(4.15.4) \quad \mathbb{V}^\circ := \left(\bigcap_{Q_j \text{ dominates } \mathbb{V}} U_j \right) \cap U_D$$

has the required property.

Choose $v \in \mathbb{V}^\circ(k)$. The line $k \cdot v$ is not contained in $\mathbb{V} \setminus \mathbb{V}^\circ$, because it contains v ; hence its generic point belongs to $\mathbb{V}^\circ$. Now $g = v$ verifies the demand of the lemma. $\square$

4.16. Choose g as in Lemma 4.15. Let R be the strict henselization of $k[u]_{(u)}$, set $S = \operatorname{Spec} R$, write $K = \operatorname{Frac}(R)$, and put $\eta^{\operatorname{perf}} = \operatorname{Spec} K^{\operatorname{perf}}$. The natural inclusion $k \rightarrow R$ gives $S \rightarrow \operatorname{Spec} k$. Let $\eta = \operatorname{Spec} K$, $s = \operatorname{Spec} k$ be the generic and special point of S , respectively. Define

$$(4.16.1) \quad \mathcal{X} = X \times_k S, \quad \mathcal{Y} = \mathbb{A}_S^1, \quad \tilde{\mathcal{F}} = \operatorname{pr}_X^* \mathcal{F},$$

$$(4.16.2) \quad \phi : \mathcal{X} \rightarrow \mathcal{Y}, \quad \phi(x, u) = f(x) + ug(x),$$

where $\operatorname{pr}_X : \mathcal{X} \rightarrow X$ is the projection. The differential $d\phi$ gives a section

$$(4.16.3) \quad \sigma = d\phi = df + u dg : \mathcal{X} \longrightarrow T^*(\mathcal{X}/S) = T^*X \times_k S.$$

Put $\mathcal{W} = \sigma^{-1}(SS(\mathcal{F}) \times_k S)$, then by Lemma 4.15,

$$(4.16.4) \quad \mathcal{W}_s = Z = Z_{X,f}, \quad \mathcal{W}_\eta \text{ is finite over } K,$$

and hence $\mathcal{W}_{\eta^{\operatorname{perf}}}$ is finite over K^{perf} .

For $t \in \{\eta, s\}$, ϕ_t is $(SS(\mathcal{F}) \times_k S)_t$ -transversal outside $\mathcal{W}_t$. For $t = s$, [13, Lemma 4.2(5)] implies that ϕ_s is universally locally acyclic relatively to $\tilde{\mathcal{F}}_s$ outside $\mathcal{W}_s$. The projection

$$(4.16.5) \quad \operatorname{pr}_S : \mathcal{X} = X \times_k S \rightarrow S$$

is universally locally acyclic relatively to $\tilde{\mathcal{F}}$, and

$$(4.16.6) \quad R\Psi_{\operatorname{pr}_S}(\tilde{\mathcal{F}}_\eta) \simeq \tilde{\mathcal{F}}_s \simeq \mathcal{F}.$$

4.17. Retain the notation of §4.16. Initially, the generic fiber is over K which is not perfect in general; however, the construction of characteristic cycle in [13] needs the base field to be perfect. In the following, we build a descent result to pass to the perfect closure of K .

Lemma 4.18. *Assume that k is algebraically closed. Let X be a smooth and purely d -dimensional scheme over k , let $\mathcal{F} \in D_{\operatorname{ctf}}(X, \Lambda)$, and let K^{perf} be the perfect closure of K . Then we have*

$$(4.18.1) \quad SS(\tilde{\mathcal{F}}_{\eta^{\operatorname{perf}}}) = SS(\mathcal{F}) \times_k \eta^{\operatorname{perf}}, \quad CC(\tilde{\mathcal{F}}_{\eta^{\operatorname{perf}}}) = CC(\mathcal{F}) \times_k \eta^{\operatorname{perf}},$$

where the second term is the flat pullback of cycle.

Proof. Under the canonical identification

$$(4.18.2) \quad T^*(\mathcal{X}_{\eta^{\operatorname{perf}}}/K^{\operatorname{perf}}) \simeq T^*(X/k) \times_k \eta^{\operatorname{perf}},$$

the complex $\tilde{\mathcal{F}}_{\eta^{\operatorname{perf}}}$ is the scalar extension of $\mathcal{F}$. Hence by [4, Theorem 1.4(iii)] we have

$$(4.18.3) \quad SS(\tilde{\mathcal{F}}_{\eta^{\operatorname{perf}}}) = SS(\mathcal{F}) \times_k \eta^{\operatorname{perf}}.$$

Put

$$(4.18.4) \quad C = SS(\mathcal{F}) = \bigcup_{\alpha} C_{\alpha}, \quad CC(\mathcal{F}) = \sum_{\alpha} m_{\alpha} [C_{\alpha}],$$

where the C_{α} are all the irreducible components of C , allowing $m_{\alpha} = 0$, and set

$$(4.18.5) \quad C_{\operatorname{perf}} = C \times_k \eta^{\operatorname{perf}}, \quad B = CC(\mathcal{F}) \times_k \eta^{\operatorname{perf}}.$$

Every irreducible component of C_{perf} has dimension d , and B is supported on C_{perf} . It suffices to verify the Milnor formula for B .

Let

$$(4.18.6) \quad j : U \longrightarrow \mathcal{X}_{\eta^{\operatorname{perf}}}, \quad f : U \longrightarrow Y$$

be a test pair with an isolated characteristic point, i.e., j is étale, Y is a smooth curve over K^{perf} , and $u_0 \in U$ is a closed point, with a geometric point $\bar{u}$ above it, which is at most an isolated $j^*C_{\operatorname{perf}}$ -characteristic point of f .

If f is j^*C_{perf} -transversal at u_0 , then we may shrink U around u_0 . The local intersection number is zero and [13, Lemma 4.2(5)] gives

$$(4.18.7) \quad R\Phi_{\tilde{u}}(j^*\tilde{\mathcal{F}}_{\eta^{\text{perf}}}, f) = 0.$$

Thus the Milnor formula holds in this case.

Assume henceforth that u_0 is isolated characteristic point. In this case, we may assume $Y = \mathbb{A}_{K^{\text{perf}}}^1$ by shrinking around u_0 and passing to an étale cover. Write

$$(4.18.8) \quad df : U \longrightarrow T^*(U/K^{\text{perf}})$$

for the section defined by the relative differential of f . Since j is étale, we may regard j^*C_{perf} as a closed conical subset of $T^*(U/K^{\text{perf}})$, endowed with its reduced scheme structure. After shrinking U once more, assume that the scheme-theoretic characteristic locus

$$(4.18.9) \quad Z_{\text{perf}} := U \times_{df, T^*(U/K^{\text{perf}})} j^*C_{\text{perf}}$$

is finite over K^{perf} .

The field K is algebraic over $k(u)$, since it is the fraction field of the strict henselization of $k[u]_{(u)}$, and K^{perf}/K is purely inseparable. Hence $K^{\text{perf}}/k(u)$ is algebraic. Since all data in the test pair are of finite presentation, they descend to a finitely generated integral $k[u]$ -subalgebra $A_0 \subset K^{\text{perf}}$. The extension $\text{Frac}(A_0)/k(u)$ is finite. Since k is perfect, we may find a localization of A_0 , denoted by A , such that $A \subset K^{\text{perf}}$ is a finitely generated smooth integral k -algebra containing $k[u]$ and carrying the descended data. Put $M = \text{Spec } A$, $F_0 = \text{Frac}(A)$. Thus A is of finite type over k , and K^{perf}/F_0 is algebraic. After enlarging the model if necessary, we obtain an étale morphism and an M -morphism

$$(4.18.10) \quad j_A : U_A \longrightarrow X \times_k M, \quad f_A : U_A \longrightarrow \mathbb{A}_M^1$$

whose base change to η^{perf} is the given test. Put

$$(4.18.11) \quad h = \text{pr}_X \circ j_A : U_A \rightarrow X, \quad \pi = \text{pr}_M \circ j_A : U_A \rightarrow M, \quad \mathcal{F}_A = j_A^* \text{pr}_X^* \mathcal{F},$$

and, using the cotangent-bundle isomorphism induced by j_A , put

$$(4.18.12) \quad C_A = U_A \times_X C \subset T^*(U_A/M),$$

$$(4.18.13) \quad B_A = j_A^*(CC(\mathcal{F}) \times_k M) = \sum_{\alpha} m_{\alpha} [U_A \times_X C_{\alpha}].$$

Let $df_A : U_A \longrightarrow T^*(U_A/M)$ be the section defined by the relative differential of f_A , and define

$$(4.18.14) \quad Z_A = U_A \times_{df_A, T^*(U_A/M)} C_A.$$

We have $(Z_A)_{\eta^{\text{perf}}} = Z_{\text{perf}}$.

The generic fiber $(Z_A)_{F_0}$ is finite, since it becomes the finite scheme Z_{perf} after the faithfully flat extension $F_0 \subset K^{\text{perf}}$. The open quasi-finite locus of $Z_A \rightarrow M$ therefore contains the generic fiber. Its complement has only finitely many irreducible components, none of which dominates M . After removing the closures of their images from M , we may assume that $Z_A \rightarrow M$ is quasi-finite.

Since $j_A = (h, \pi)$ is étale, the morphism h is smooth. Hence h is C -transversal and $SS(\mathcal{F}_A) = h^*C$ by [13, Lemma 3.4(1) and Corollary 4.5(2)]. Under the cotangent decomposition induced by j_A , this equality becomes

$$(4.18.15) \quad T^*(U_A/k) \simeq T^*(U_A/M) \oplus \pi^*T^*(M/k), \quad h^*C = C_A \oplus 0.$$

For the M -morphism f_A , the composition of its cotangent map

$$(4.18.16) \quad df_A : T^*(\mathbb{A}_M^1/k) \times_{\mathbb{A}_M^1} U_A \rightarrow T^*(U_A/k)$$

with the projection $T^*(U_A/k) \rightarrow T^*(U_A/M)$ sends (adt, β) to adf_A , while covectors pulled back from M map identically to the second summand $\pi^*T^*(M/k)$. Consequently, by the definition of transversality and the conicality of C_A , the non- C -transversal locus of (h, f_A) is precisely the underlying set of $(df_A)^{-1}(C_A) = Z_A$. Thus (h, f_A) is C -transversal on $U_A \setminus Z_A$. Since the cotangent map of π has image in the second summand $\pi^*T^*(M/k)$, which meets $C_A \oplus 0$ only in the zero-section, the pair (h, π) is C -transversal everywhere.

For a geometric point $\bar{z}$ of Z_A , with image $\bar{s}$ in M , define

$$(4.18.17) \quad \begin{aligned} I(\bar{z}) &= (B_{A, \bar{s}}, df_{A, \bar{s}})_{T^*(U_{A, \bar{s}}/\bar{s}), \bar{z}}, \\ \phi(\bar{z}) &= -\dim_{\text{tot}} R\Phi_{\bar{z}}(\mathcal{F}_{A, \bar{s}}, f_{A, \bar{s}}). \end{aligned}$$

For every α , set

$$(4.18.18) \quad \mathcal{A}_{\alpha, A} = \mathcal{O}_{U_A \times_X C_\alpha} \otimes^{\mathbf{L}}_{\mathcal{O}_{T^*(U_A/M)} df_A} \mathcal{O}_{U_A},$$

where the last module structure is defined by the section df_A . Since M is smooth over k and U_A is smooth over M , the scheme $T^*(U_A/M)$ is regular. Hence $\mathcal{O}_{U_A \times_X C_\alpha}$ is perfect on it, and $\mathcal{A}_{\alpha, A}$ is perfect on U_A . Since $U_A \rightarrow M$ is smooth, $\mathcal{A}_{\alpha, A}$ has finite Tor-dimension over M . Moreover,

$$(4.18.19) \quad U_A \times_X C_\alpha = U_A \times_{X \times_k M} (C_\alpha \times_k M)$$

is étale over $C_\alpha \times_k M$, and therefore is flat over M ; the scheme U_A is also flat over M . By derived base change we have

$$(4.18.20) \quad (\mathcal{A}_{\alpha, A})_{\bar{s}} = Li_{\bar{s}}^* \mathcal{A}_{\alpha, A} \simeq \mathcal{O}_{U_{A, \bar{s}} \times_X C_\alpha} \otimes^{\mathbf{L}}_{\mathcal{O}_{T^*(U_{A, \bar{s}}/\bar{s}), df_{A, \bar{s}}}} \mathcal{O}_{U_{A, \bar{s}}}$$

where $\bar{s} \rightarrow M$ is any geometric point. Thus $(\mathcal{A}_{\alpha, A})_{\bar{s}}$ is the derived local intersection of $U_{A, \bar{s}} \times_X C_\alpha$ and $df_{A, \bar{s}}(U_{A, \bar{s}})$ in $T^*(U_{A, \bar{s}}/\bar{s})$. Its local Euler characteristic is the corresponding isolated intersection multiplicity in the definition of the function I . The cohomology of $\mathcal{A}_{\alpha, A}$ is supported on

$$(4.18.21) \quad (U_A \times_X C_\alpha) \cap df_A(U_A) \subset Z_A,$$

hence on a closed subscheme quasi-finite over M . Therefore [13, Lemma 2.3 and the proof of Proposition 5.8], followed by summing with the coefficients m_α , shows that I is constructible and flat over M in the sense of [13, Definition 2.1].

The transversality of (h, π) and of (h, f_A) outside Z_A , together with [13, Lemma 4.2(5)], shows that π is universally locally acyclic relatively to $\mathcal{F}_A$ and that f_A is universally locally acyclic relatively to $\mathcal{F}_A$ on $U_A \setminus Z_A$. By [13, Proposition 2.16] we have ϕ is also constructible and flat over M since $\mathbb{A}_M^1 \rightarrow M$ is a smooth relative curve and $Z_A \rightarrow M$ is quasi-finite.

Let t be a closed point of M . Since A is of finite type over the algebraically closed field k , one has $\kappa(t) = k$, and $h_t : U_{A, t} \rightarrow X$ is étale. By [13, Lemma 5.11(2)], we have

$$(4.18.22) \quad B_{A, t} = h_t^* CC(\mathcal{F}) = CC(h_t^* \mathcal{F}) = CC(\mathcal{F}_{A, t}).$$

The fiber $Z_{A, t}$ is finite. Saito's Milnor formula [13, Theorem 5.9] therefore gives

$$(4.18.23) \quad I(\bar{z}) = \phi(\bar{z})$$

for every geometric point $\bar{z}$ above every closed point $t \in M$.

Notice that $I - \phi$ is also a constructible function. On every irreducible component Z' of Z_A , constructibility gives a dense open subset V on which $I - \phi$ equals its value at the geometric generic point of Z' . Since Z' is of finite type over the algebraically closed field k , it is Jacobson, and V contains a closed k -point z ; its image in M , z_M , is also closed. Hence regard $\bar{z}$ as a geometric point of Z_A lying above z_M , by (4.18.23) we have $I(\bar{z}) = \phi(\bar{z})$. Thus $I - \phi$ vanishes at the generic point of every irreducible component of Z_A . Saito [13, Lemma 2.2(3)] now gives $I - \phi = 0$ on all geometric points of Z_A .

Choose an algebraic closure Ω of K^{perf} together with an embedding $\kappa(u_0) \hookrightarrow \Omega$ denoted by $\bar{u}$. Since K^{perf}/F_0 is algebraic, Ω is also an algebraic closure of F_0 , so $\bar{u}$ defines a geometric point of Z_A above the geometric generic point of M . Moreover, $\kappa(u_0)/K^{\text{perf}}$ is finite separable because K^{perf} is perfect. Flat base change therefore preserves the local intersection multiplicity at the chosen geometric point, hence we have

$$(4.18.24) \quad I(\bar{u}) = (j^*B, df)_{T^*(U/K^{\text{perf}}), u_0}.$$

Since $Z_A \rightarrow M$ is quasi-finite, the induced morphism $Z_A \rightarrow \mathbb{A}_M^1$ is also quasi-finite. By the universal local acyclicity of f_A outside Z_A and [13, Proposition 2.8(1)], we may identify the vanishing-cycle complex defining $\phi(\bar{u})$ with $R\Phi_{\bar{u}}(j^*\tilde{\mathcal{F}}_{\eta^{\text{perf}}}, f)$. Consequently,

$$(4.18.25) \quad \phi(\bar{u}) = -\dim_{\text{tot}} R\Phi_{\bar{u}}(j^*\tilde{\mathcal{F}}_{\eta^{\text{perf}}}, f).$$

Evaluating $I - \phi = 0$ at $\bar{u}$ yields

$$(4.18.26) \quad (j^*B, df)_{T^*(U/K^{\text{perf}}), u_0} = -\dim_{\text{tot}} R\Phi_{\bar{u}}(j^*\tilde{\mathcal{F}}_{\eta^{\text{perf}}}, f).$$

Thus B satisfies the Milnor formula for every at most isolated C_{perf} -characteristic test. Since $C_{\text{perf}} = SS(\tilde{\mathcal{F}}_{\eta^{\text{perf}}})$, by the uniqueness in [13, Theorem 5.9] we have

$$(4.18.27) \quad CC(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}) = CC(\mathcal{F}) \times_k \eta^{\text{perf}}.$$

This proves (4.18.1). □

4.19. We now verify the local-acyclicity condition on the ordinary generic fiber. By (4.18.1),

$$(4.19.1) \quad SS(\mathcal{F}) \times_k \eta^{\text{perf}} = SS(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}).$$

Since $\phi_{\eta^{\text{perf}}}$ is $SS(\mathcal{F})_{\eta^{\text{perf}}}$ -transversal outside $\mathcal{W}_{\eta^{\text{perf}}}$, [13, Lemma 4.2(5)] implies that $\phi_{\eta^{\text{perf}}}$ is universally locally acyclic relatively to $\tilde{\mathcal{F}}_{\eta^{\text{perf}}}$ outside $\mathcal{W}_{\eta^{\text{perf}}}$. Notice that the morphism $\eta^{\text{perf}} = \text{Spec } K^{\text{perf}} \rightarrow \text{Spec } K = \eta$ is a universal homeomorphism. The induced equivalences of étale topoi [15, Tag 03SI], together with base change identify the canonical morphisms in Lemma 2.2(4) before and after this extension. Hence universal local acyclicity descends by the equivalence of (2) and (4) in Lemma 2.2. Therefore, we have

$$(4.19.2) \quad \phi_{\eta} : \mathcal{X}_{\eta} \longrightarrow \mathcal{Y}_{\eta}$$

is universally locally acyclic relatively to $\tilde{\mathcal{F}}_{\eta}$ outside $\mathcal{W}_{\eta}$. By a slight abuse of notation, we denote both the specialization map on $H^0(\mathcal{X}_{\eta}, \mathcal{K}_{\mathcal{X}_{\eta}/\eta})$ and $H^0(\mathcal{X}_{\eta}, \mathcal{K}_{\mathcal{X}_{\eta}/\mathcal{Y}_{\eta}/\eta})$ by $\text{sp}^{\text{ét}}$. The universal local acyclicity of $\text{pr}_S : \mathcal{X} \rightarrow S$ gives

$$(4.19.3) \quad R\Psi_{\text{pr}_S}(\tilde{\mathcal{F}}_{\eta}) \simeq \tilde{\mathcal{F}}_s \simeq \mathcal{F},$$

therefore Proposition 3.3(6) gives

$$(4.19.4) \quad \text{sp}^{\text{ét}} \tilde{C}_{\mathcal{X}_{\eta}/\mathcal{Y}_{\eta}/\eta}^{\mathcal{W}_{\eta}}(\tilde{\mathcal{F}}_{\eta}) = \tilde{C}_{X/\mathbb{A}_k^1/k}^Z(\mathcal{F}).$$

4.20. Recall that we have the section $\sigma = d\phi = df + u dg$. Combining Lemma 4.18 with flat base change for the refined intersection, we have

$$(4.20.1) \quad \sigma_{\eta}^!(CC(\mathcal{F}) \times_k \eta) \times_{\eta} \eta^{\text{perf}} = \sigma_{\eta^{\text{perf}}}^!(CC(\mathcal{F}) \times_k \eta^{\text{perf}}) = \sigma_{\eta^{\text{perf}}}^! CC(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}).$$

Lemma 4.21. *With the preceding notation, we have*

$$(4.21.1) \quad \tilde{C}_{\mathcal{X}_{\eta^{\text{perf}}}/\mathcal{Y}_{\eta^{\text{perf}}}/\eta^{\text{perf}}}^{\mathcal{W}_{\eta^{\text{perf}}}}(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}) = \tilde{\text{cl}}\left(\sigma_{\eta^{\text{perf}}}^!(CC(\mathcal{F}) \times_k \eta^{\text{perf}})\right).$$

Proof. Recall that under our construction, $\mathcal{W}_{\eta^{\text{perf}}}$ is finite over η^{perf} . For each closed point $x \in |\mathcal{W}_{\eta^{\text{perf}}}|$, choose an open neighbourhood U_x containing no other point of $|\mathcal{W}_{\eta^{\text{perf}}}|$. The étale locality and the cohomological Milnor formula of non-acyclicity class [18, Proposition 4.6 and Theorem 6.1] give

$$(4.21.2) \quad C_{U_x/\mathcal{Y}_{\eta^{\text{perf}}}/\eta^{\text{perf}}}^{\{x\}}(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}^{\text{perf}}|_{U_x}) = -\dim_{\text{tot}} R\Phi_{\tilde{x}}(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}^{\text{perf}}, \phi_{\eta^{\text{perf}}}) \quad \text{in } H_x^0(U_x, \mathcal{K}_{U_x/\eta^{\text{perf}}}) \simeq \Lambda.$$

Saito's isolated characteristic-cycle formula [13, Theorem 5.9] says, for the same point,

$$(4.21.3) \quad \begin{aligned} \text{mult}_x(\sigma_{\eta^{\text{perf}}}^! CC(\tilde{\mathcal{F}}_{\eta^{\text{perf}}})) \\ = (CC(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}), d\phi_{\eta^{\text{perf}}})_{T^*(\mathcal{X}_{\eta^{\text{perf}}}/\eta^{\text{perf}}), x} = -\dim_{\text{tot}} R\Phi_{\tilde{x}}(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}^{\text{perf}}, \phi_{\eta^{\text{perf}}}) \quad \text{in } \mathbf{Z}. \end{aligned}$$

Since supported cohomology on a finite set decomposes as the direct sum of its point-supported cohomology, summing over x gives

$$(4.21.4) \quad \tilde{C}_{\mathcal{X}_{\eta^{\text{perf}}}/\mathcal{Y}_{\eta^{\text{perf}}}/\eta^{\text{perf}}}^{\mathcal{W}_{\eta^{\text{perf}}}}(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}) = \tilde{\text{cl}}(\sigma_{\eta^{\text{perf}}}^! CC(\tilde{\mathcal{F}}_{\eta^{\text{perf}}})) .$$

Applying (4.20.1) we get the desired result. $\square$

Lemma 4.22. *Let $q_X : \mathcal{X}_{\eta^{\text{perf}}} \rightarrow \mathcal{X}_{\eta}$ and $q_Y : \mathcal{Y}_{\eta^{\text{perf}}} \rightarrow \mathcal{Y}_{\eta}$ be the projections. Then there is a canonical isomorphism*

$$(4.22.1) \quad q_X^* \mathcal{K}_{\mathcal{X}_{\eta}/\mathcal{Y}_{\eta}/\eta} \xrightarrow{\sim} \mathcal{K}_{\mathcal{X}_{\eta^{\text{perf}}}/\mathcal{Y}_{\eta^{\text{perf}}}/\eta^{\text{perf}}} .$$

Moreover, we have

$$(4.22.2) \quad q_X^* : H_{\mathcal{W}_{\eta}}^0(\mathcal{X}_{\eta}, \mathcal{K}_{\mathcal{X}_{\eta}/\mathcal{Y}_{\eta}/\eta}) \xrightarrow{\sim} H_{\mathcal{W}_{\eta^{\text{perf}}}}^0(\mathcal{X}_{\eta^{\text{perf}}}, \mathcal{K}_{\mathcal{X}_{\eta^{\text{perf}}}/\mathcal{Y}_{\eta^{\text{perf}}}/\eta^{\text{perf}}}) .$$

Proof. The maps q_X and q_Y are universal homeomorphisms, hence q_X^* and q_Y^* are equivalences on the relevant derived categories of constructible sheaves by [3, Exposé VIII, Théorème 1.1]. For the Cartesian square defined by $\phi_{\eta} : \mathcal{X}_{\eta} \rightarrow \mathcal{Y}_{\eta}$, base change gives

$$(4.22.3) \quad q_Y^* R\phi_{\eta!} \xrightarrow{\sim} R\phi_{\eta^{\text{perf}}!} q_X^* ,$$

by [15, Tag 0F7L]. Moreover, the two equivalences yield a canonical base change isomorphism

$$(4.22.4) \quad q_X^* R\phi_{\eta}^! \xrightarrow{\sim} R\phi_{\eta^{\text{perf}}}^! q_Y^* .$$

Hence by definition we have

$$(4.22.5) \quad q_X^* \mathcal{K}_{\mathcal{X}_{\eta}/\eta} \xrightarrow{\sim} \mathcal{K}_{\mathcal{X}_{\eta^{\text{perf}}}/\eta^{\text{perf}}}, \quad q_X^* \mathcal{K}_{\mathcal{X}_{\eta}/\mathcal{Y}_{\eta}} \xrightarrow{\sim} \mathcal{K}_{\mathcal{X}_{\eta^{\text{perf}}}/\mathcal{Y}_{\eta^{\text{perf}}}},$$

and the consequence follows from the distinguished triangle (3.2.3). $\square$

Lemma 4.23. *Retain the preceding notation. We have*

$$(4.23.1) \quad \tilde{C}_{\mathcal{X}_{\eta}/\mathcal{Y}_{\eta}/\eta}^{\mathcal{W}_{\eta}}(\tilde{\mathcal{F}}_{\eta}) = \tilde{\text{cl}}(\sigma_{\eta}^! (CC(\mathcal{F}) \times_k \eta)) .$$

Proof. Let q_X and q_Y be as before. The distinguished triangle (3.2.3) gives

$$(4.23.2) \quad \rho_s : H^0(X, \mathcal{K}_{X/k}) \rightarrow H^0(X, \mathcal{K}_{X/\mathbb{A}_k^1/k}), \quad \rho_{\eta} : H^0(\mathcal{X}_{\eta}, \mathcal{K}_{\mathcal{X}_{\eta}/\eta}) \rightarrow H^0(\mathcal{X}_{\eta}, \mathcal{K}_{\mathcal{X}_{\eta}/\mathcal{Y}_{\eta}/\eta}),$$

and similarly we define $\rho_{\eta^{\text{perf}}}$. Thus

$$(4.23.3) \quad \tilde{\text{cl}}_{\eta} = \rho_{\eta} \circ \text{cl}_{\eta}, \quad \tilde{\text{cl}}_{\eta^{\text{perf}}} = \rho_{\eta^{\text{perf}}} \circ \text{cl}_{\eta^{\text{perf}}} .$$

We compare the pullbacks of the two sides of (4.23.1). Firstly, the pullback formula for the non-acyclicity class (3.3.2) gives

$$(4.23.4) \quad q_X^* \tilde{C}_{\mathcal{X}_{\eta}/\mathcal{Y}_{\eta}/\eta}^{\mathcal{W}_{\eta}}(\tilde{\mathcal{F}}_{\eta}) = \tilde{C}_{\mathcal{X}_{\eta^{\text{perf}}}/\mathcal{Y}_{\eta^{\text{perf}}}/\eta^{\text{perf}}}^{\mathcal{W}_{\eta^{\text{perf}}}}(\tilde{\mathcal{F}}_{\eta^{\text{perf}}}) .$$

On the cycle side, compatibility of the supported cycle class map with scalar extension [7, Cycle, Lemme 2.3.7] gives

$$(4.23.5) \quad q_X^* \text{cl}_\eta(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)) = \text{cl}_{\eta^{\text{perf}}}(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta) \times_\eta \eta^{\text{perf}}).$$

Moreover, the naturality of morphisms involved gives the commutative square

$$(4.23.6) \quad \begin{array}{ccc} H_{\mathcal{W}_\eta}^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\eta}) & \xrightarrow{\rho_\eta} & H_{\mathcal{W}_\eta}^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}) \\ q_X^* \downarrow & & \downarrow q_X^* \\ H_{\mathcal{W}_{\eta^{\text{perf}}}}^0(\mathcal{X}_{\eta^{\text{perf}}}, \mathcal{K}_{\mathcal{X}_{\eta^{\text{perf}}}/\eta^{\text{perf}}}) & \xrightarrow{\rho_{\eta^{\text{perf}}}} & H_{\mathcal{W}_{\eta^{\text{perf}}}}^0(\mathcal{X}_{\eta^{\text{perf}}}, \mathcal{K}_{\mathcal{X}_{\eta^{\text{perf}}}/\mathcal{Y}_{\eta^{\text{perf}}}/\eta^{\text{perf}}}). \end{array}$$

Consequently, using (4.20.1), we have

$$(4.23.7) \quad \begin{aligned} q_X^* \tilde{\text{cl}}_\eta(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)) &= \tilde{\text{cl}}_{\eta^{\text{perf}}}(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta) \times_\eta \eta^{\text{perf}}) \\ &= \tilde{\text{cl}}_{\eta^{\text{perf}}}(\sigma_{\eta^{\text{perf}}}^!(CC(\mathcal{F}) \times_k \eta^{\text{perf}})). \end{aligned}$$

By (4.21.1), the right-hand sides of (4.23.4) and (4.23.7) are equal, hence

$$(4.23.8) \quad q_X^* \tilde{C}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}^{\mathcal{W}_\eta}(\tilde{\mathcal{F}}_\eta) = q_X^* \tilde{\text{cl}}_\eta(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)).$$

Finally, by the previous lemma 4.22, the equality descends uniquely to η , and we obtain

$$(4.23.9) \quad \tilde{C}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}^{\mathcal{W}_\eta}(\tilde{\mathcal{F}}_\eta) = \tilde{\text{cl}}_\eta(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta))$$

which is (4.23.1). □

4.24. *Proof of Theorem 4.2.* Recall that

$$(4.24.1) \quad \rho_s : H^0(X, \mathcal{K}_{X/k}) \rightarrow H^0(X, \mathcal{K}_{X/\mathbb{A}_k^1/k}), \quad \rho_\eta : H^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\eta}) \rightarrow H^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta})$$

and by definition we have $\tilde{\text{cl}}_s = \rho_s \circ \text{cl}_s$, $\tilde{\text{cl}}_\eta = \rho_\eta \circ \text{cl}_\eta$. By [18, (4.47)], we have

$$(4.24.2) \quad \text{sp}^{\text{ét}} \circ \rho_\eta = \rho_s \circ \text{sp}^{\text{ét}}.$$

Notice that every component of the cycle $CC(\mathcal{F}) \times_k [S] \in \text{CH}_{d+1}(SS(\mathcal{F}) \times_k S/S)$ is flat over S , Corollary 4.12 gives

$$(4.24.3) \quad \text{sp}^{\text{ét}} \text{cl}_\eta(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)) = \text{cl}_s((df)^! CC(\mathcal{F})).$$

Combining (4.19.4), (4.23.1) and (4.24.3), we have

$$(4.24.4) \quad \begin{aligned} \tilde{C}_{X/\mathbb{A}^1/k}^Z(\mathcal{F}) &= \text{sp}^{\text{ét}} \tilde{C}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}^{\mathcal{W}_\eta}(\tilde{\mathcal{F}}_\eta) \\ &= \text{sp}^{\text{ét}} \rho_\eta \text{cl}_\eta(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)) \\ &= \rho_s \text{sp}^{\text{ét}} \text{cl}_s(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)) \\ &= \rho_s \text{cl}_s((df)^! CC(\mathcal{F})). \end{aligned}$$

This is Theorem 4.2. □

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